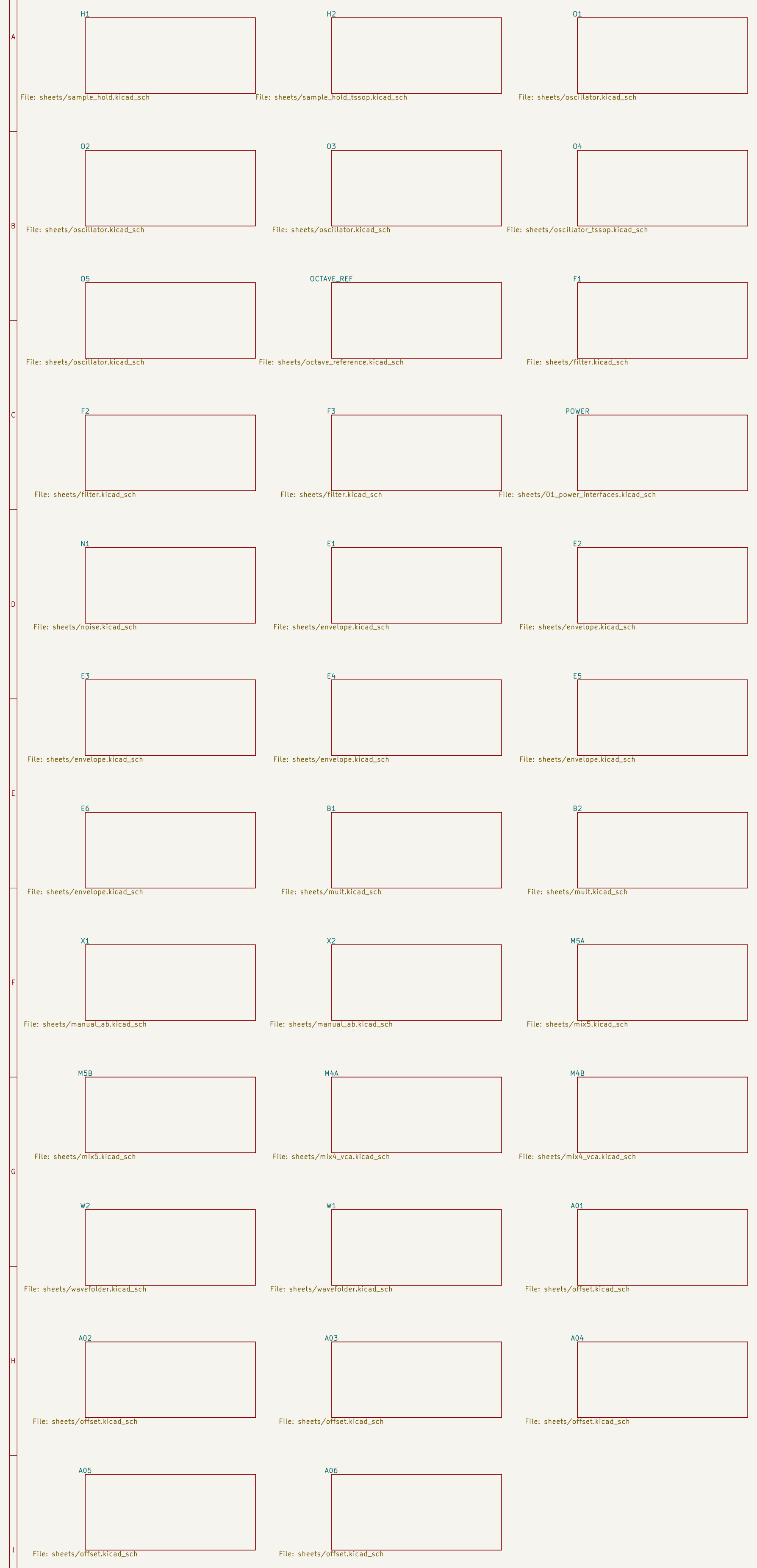
H1
File: sheets/sample_hold.kicad_sch
H2
File: sheets/sample_hold_tssop.kicad_sch
O1
File: sheets/oscillator.kicad_sch
O2
File: sheets/oscillator.kicad_sch
O3
File: sheets/oscillator.kicad_sch
O4
File: sheets/oscillator_tssop.kicad_sch
O5
File: sheets/oscillator.kicad_sch
OCTAVE_REF
File: sheets/octave_reference.kicad_sch
F1
File: sheets/filter.kicad_sch
F2
File: sheets/filter.kicad_sch
F3
File: sheets/filter.kicad_sch
POWER
File: sheets/01_power_interfaces.kicad_sch
N1
File: sheets/noise.kicad_sch
E1
File: sheets/envelope.kicad_sch
E2
File: sheets/envelope.kicad_sch
E3
File: sheets/envelope.kicad_sch
E4
File: sheets/envelope.kicad_sch
E5
File: sheets/envelope.kicad_sch
E6
File: sheets/envelope.kicad_sch
B1
File: sheets/mult.kicad_sch
B2
File: sheets/mult.kicad_sch
X1
File: sheets/manual_ab.kicad_sch
X2
File: sheets/manual_ab.kicad_sch
M5A
File: sheets/mix5.kicad_sch
M5B
File: sheets/mix5.kicad_sch
M4A
File: sheets/mix4_vca.kicad_sch
M4B
File: sheets/mix4_vca.kicad_sch
W2
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W1
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A01
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A02
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A03
File: sheets/offset.kicad_sch
A04
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A05
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A06
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Size: A0
Date:
Rev:
KiCad E.D.A. 10.0.6
Id: 1/36

U102A
U102B
U102C
U102D
U102E
U103A
U103B
U103C
U103D
U103E
U104A
U104B
U104C
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LM393BIDR
CD74HC221M96
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DH102
DH117
BAT54S,215
BAV199,215
D509
D609
D709
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BAS16GW-QX
RV801
TRIGGER_TIP
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Cdc0c61_INV
GATE_REF
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SLEW_BUFFERED
C1979a1_DRIVE
C1979a1_FB
C1979a1_ISO_MID
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Cdded02_ENABLE
Ce0e057_SENSE
TRIGGER_BUFFERED
C267eb3_SENSE
IN_BUFFERED
IN_PROTECTED
IN_TIP
C07a7c2_SENSE
C07a7c2_DRIVE
C07a7c2_LED_A
C07a7c2_LED_K
Ca073c3_SENSE
Ca073c3_DRIVE
Ca073c3_LED_A
Ca073c3_LED_K
C307276_GATE_SENSE
C307276_THRESH
C307276_CMP_OUT
C8a1aae_DEBOUNCE
BUTTON_ACTIVE
C382003_ENABLE
C372c4e_DRIVE
C372c4e_SENSE
C372c4e_LED_A
C372c4e_LED_K
TRIGGER_GATE
TRIGGER_OR
SAMPLE_PULSE
TIMING_RCX
TIMING_C
HOLD_CAP
SLEW_POT_IN
OUT_TIP
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-12V
+5V
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magnitude_indicator_J_H1_IN_mag___A_IO_UNUSED1
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magnitude_indicator_J_H1_IN_mag___A_IO_UNUSED3
magnitude_indicator_J_H1_IN_mag___A_IO_UNUSED4
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slew_island_C_H1_SLEW___PRE_IO_UNUSED4
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10000 Ω
1e-07 F
100 nF
4.7

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Date:
Rev:
KiCad E.D.A. 10.0.6
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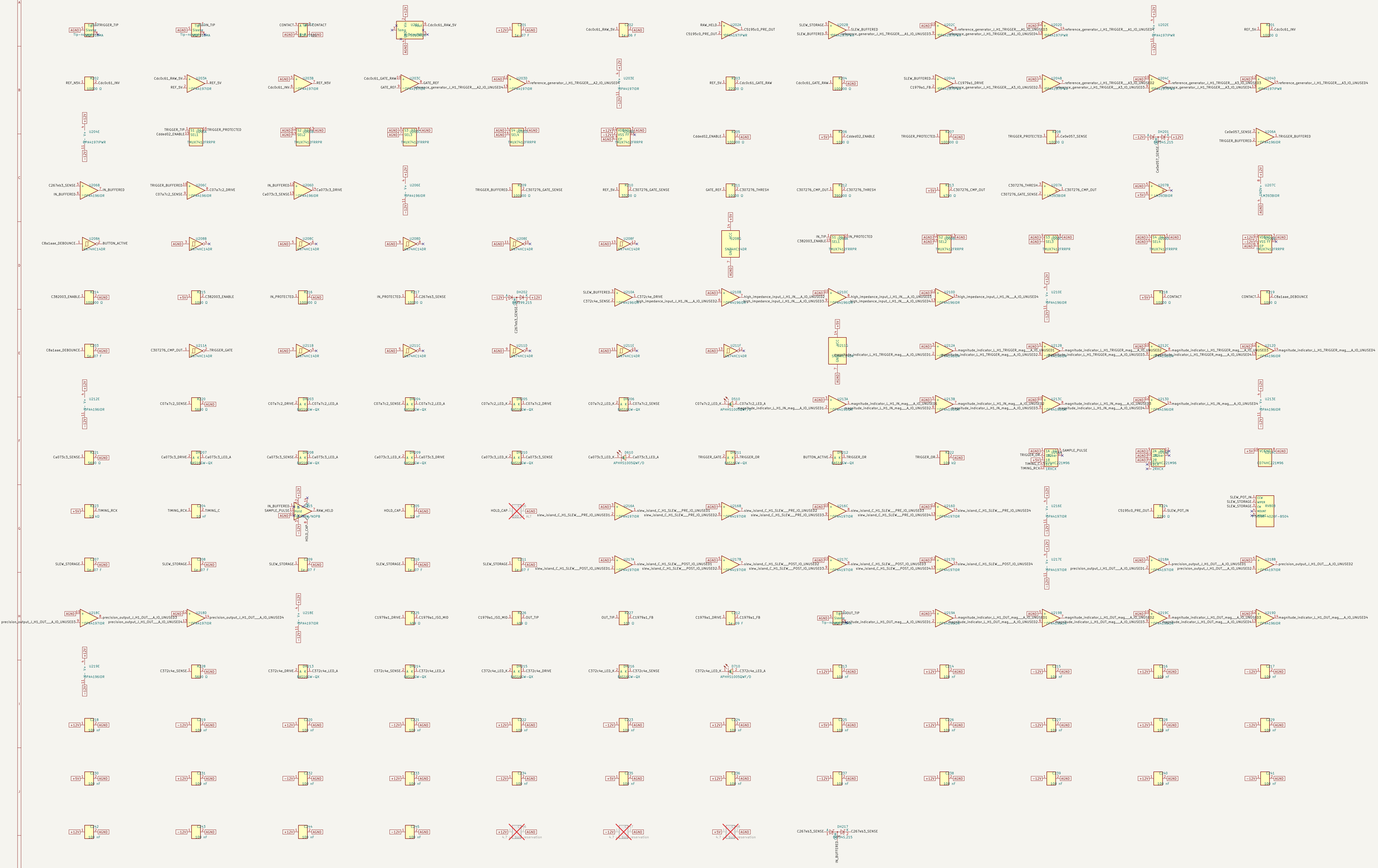

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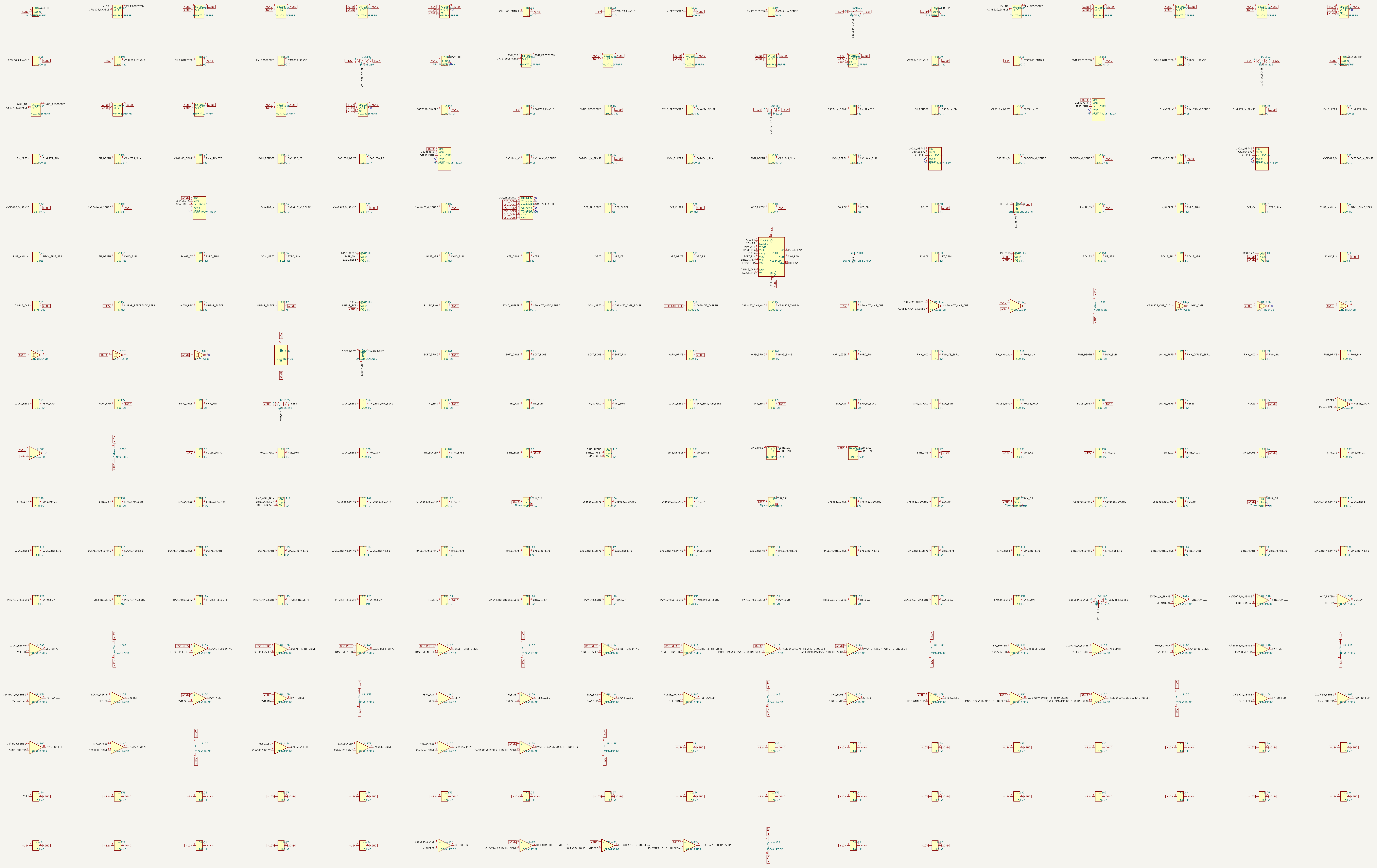
KiCad E.D.A. 10.0.6
Id: 3/36



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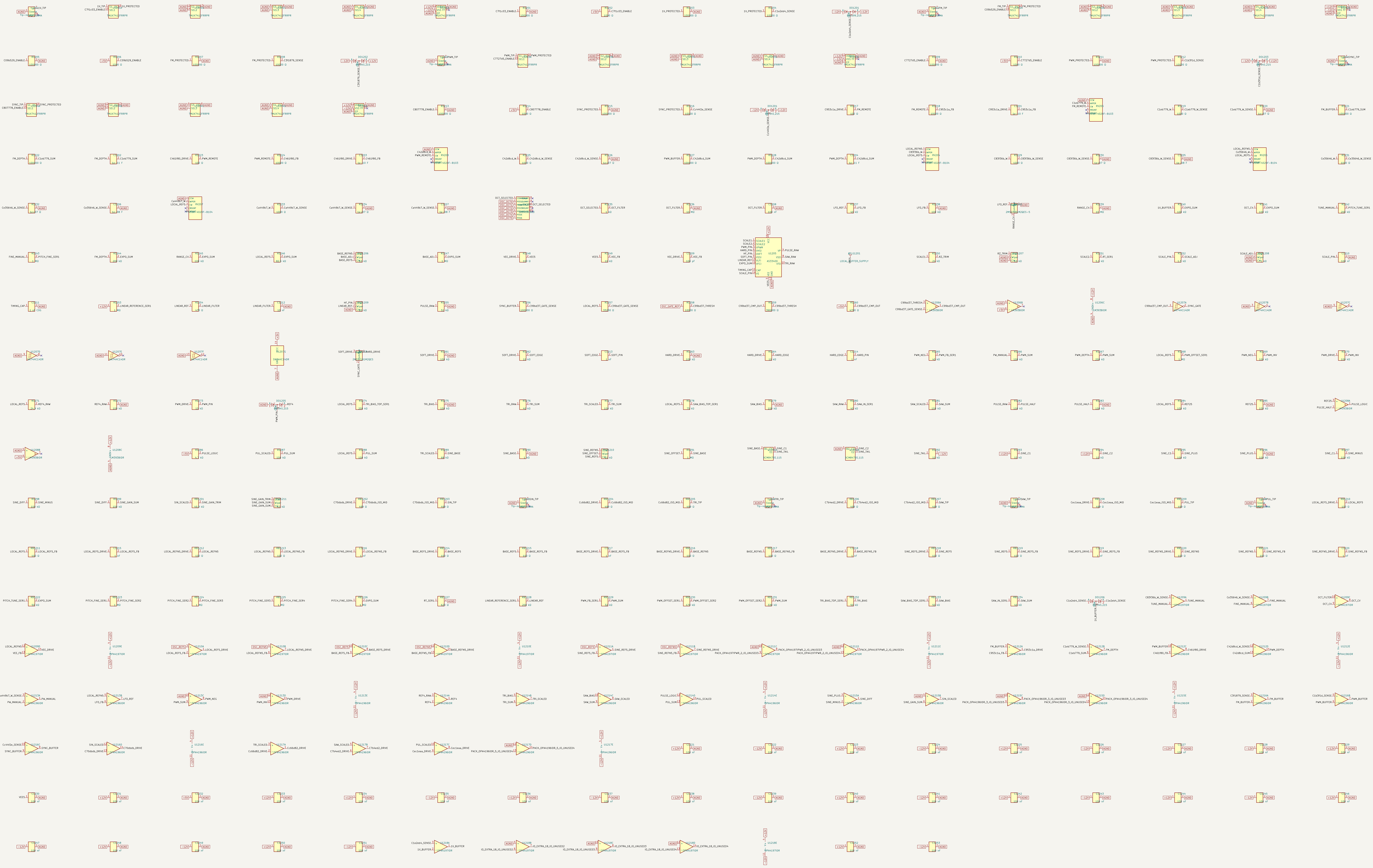
KiCad E.D.A. 10.0.6 | Id: 4/36



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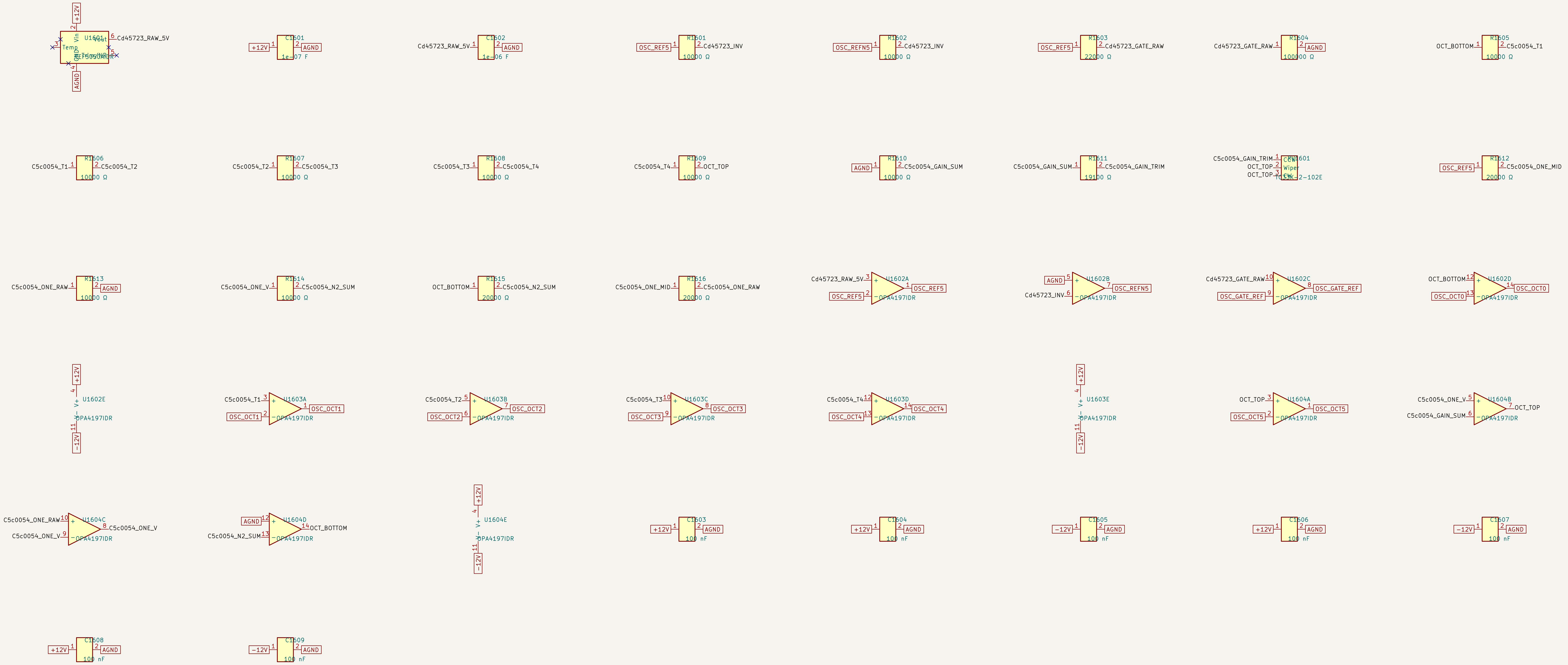
Size: A0	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 8/36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Vout
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Trim/NR
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AGND
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C1601
1e-07 F
+12V
AGND
C1602
1e-06 F
Cd45723_RAW_5V
AGND
R1601
10000 Ω
OSC_REF5
Cd45723_INV
R1602
10000 Ω
OSC_REFN5
Cd45723_INV
R1603
22000 Ω
OSC_REF5
Cd45723_GATE_RAW
R1604
100000 Ω
Cd45723_GATE_RAW
AGND
R1605
10000 Ω
OCT_BOTTOM
C5c0054_T1
R1606
10000 Ω
C5c0054_T1
C5c0054_T2
R1607
10000 Ω
C5c0054_T2
C5c0054_T3
R1608
10000 Ω
C5c0054_T3
C5c0054_T4
R1609
10000 Ω
C5c0054_T4
OCT_TOP
R1610
10000 Ω
AGND
C5c0054_GAIN_SUM
R1611
19100 Ω
C5c0054_GAIN_SUM
C5c0054_GAIN_TRIM
RV1601
TC33X-2-102E
CCW
Wiper
CW
C5c0054_GAIN_TRIM
OCT_TOP
OCT_TOP
R1612
20000 Ω
OSC_REF5
C5c0054_ONE_MID
R1613
10000 Ω
C5c0054_ONE_RAW
AGND
R1614
10000 Ω
C5c0054_ONE_V
C5c0054_N2_SUM
R1615
20000 Ω
OCT_BOTTOM
C5c0054_N2_SUM
R1616
20000 Ω
C5c0054_ONE_MID
C5c0054_ONE_RAW
U1602A
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OSC_REF5
OSC_REF5
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OPA4197IDR
AGND
Cd45723_INV
OSC_REFN5
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Cd45723_GATE_RAW
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OPA4197IDR
OCT_BOTTOM
OSC_OCT0
OSC_OCT0
U1602E
OPA4197IDR
V+
V-
+12V
-12V
U1603A
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C5c0054_T1
OSC_OCT1
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U1603B
OPA4197IDR
C5c0054_T2
OSC_OCT2
OSC_OCT2
U1603C
OPA4197IDR
C5c0054_T3
OSC_OCT3
OSC_OCT3
U1603D
OPA4197IDR
C5c0054_T4
OSC_OCT4
OSC_OCT4
U1603E
OPA4197IDR
+12V
-12V
U1604A
OPA4197IDR
OCT_TOP
OSC_OCT5
OSC_OCT5
U1604B
OPA4197IDR
C5c0054_ONE_V
C5c0054_GAIN_SUM
OCT_TOP
U1604C
OPA4197IDR
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C5c0054_ONE_V
C5c0054_ONE_V
U1604D
OPA4197IDR
AGND
C5c0054_N2_SUM
OCT_BOTTOM
U1604E
OPA4197IDR
+12V
-12V
C1603
100 nF
+12V
AGND
C1604
100 nF
+12V
AGND
C1605
100 nF
-12V
AGND
C1606
100 nF
+12V
AGND
C1607
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-12V
AGND
C1608
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AGND
C1609
100 nF
-12V
AGND

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Rev:
KiCad E.D.A. 10.0.6
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Size: A0 Date: **Rev:**
KiCad E.D.A. 10.0.6 Id: 10/36

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Size: A0 Date: **Rev:**
KiCad E.D.A. 10.0.6 Id: 11/36

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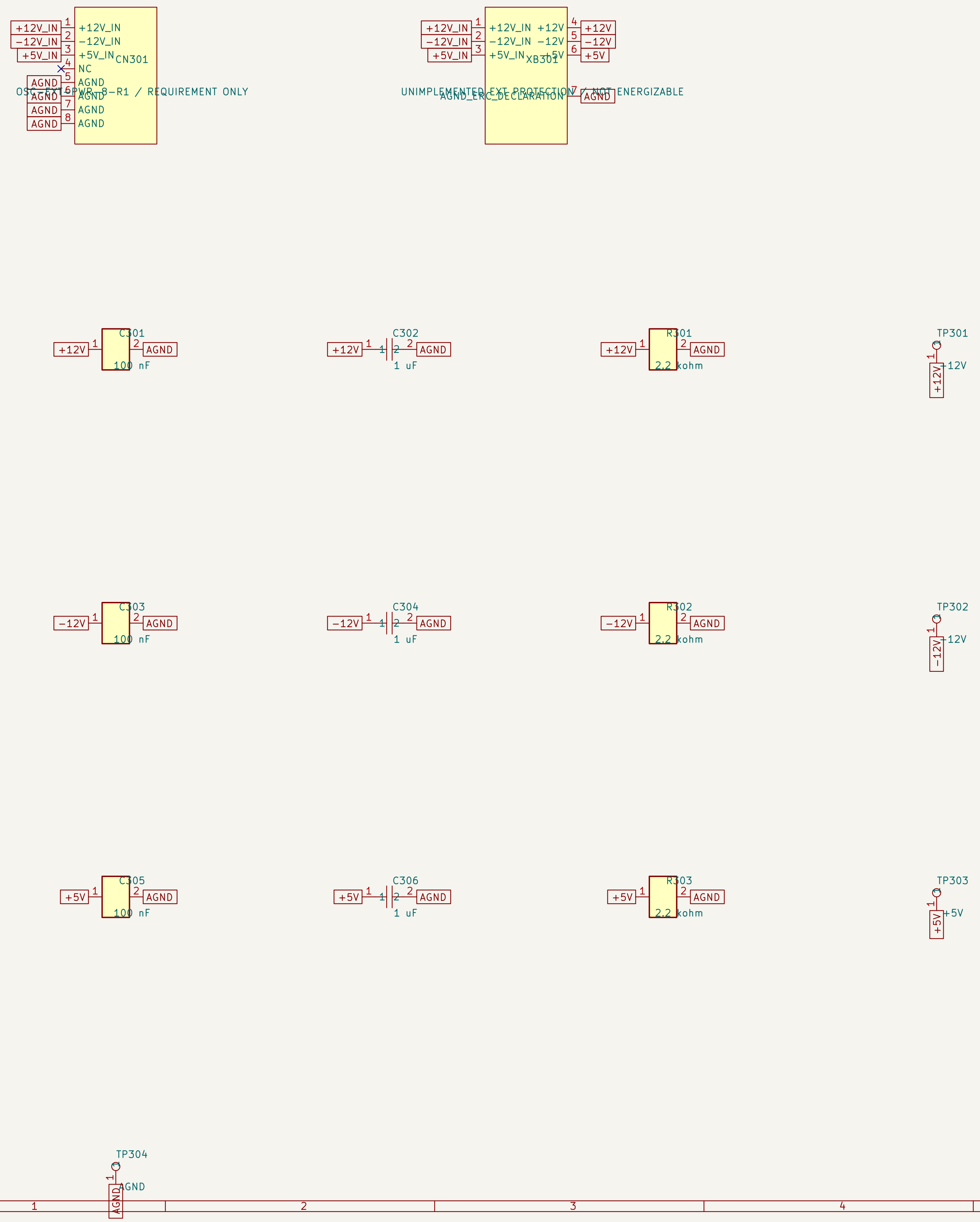
KiCad E.D.A. 10.0.6		Id: 12/36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−12V_IN
+5V_IN
NC
AGND
AGND
AGND
AGND
CN301
OSC−EXT−PWR−8−R1 / REQUIREMENT ONLY
+12V_IN
−12V_IN
+5V_IN
+12V
−12V
+5V
XB301
AGND_ERC_DECLARATION
UNIMPLEMENTED EXT PROTECTION / NOT ENERGIZABLE
AGND
C301
100 nF
+12V
AGND
C302
1 uF
R301
2.2 kohm
TP301
+12V
C303
100 nF
−12V
C304
1 uF
R302
2.2 kohm
TP302
−12V
C305
100 nF
+5V
C306
1 uF
R303
2.2 kohm
TP303
+5V
TP304
AGND

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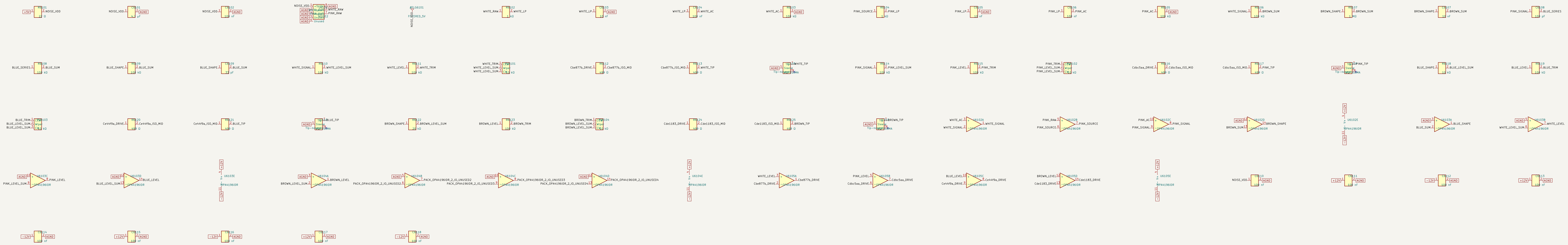
Size: A3
Date:
Rev:
KiCad E.D.A. 10.0.6
Id: 13/36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NOISE_VDD C6101 AGND 4.7 µF
NOISE_VDD C6102 AGND 100 nF
NOISE_VDD +5V0V U6101 AGND WHITE_RAW PINK_RAW NOISE2 Unused AGND
#FLG6101 NOISE_VDD FILTERED_5V
WHITE_RAW R6102 WHITE_LP 1 kΩ
WHITE_LP C6103 AGND 10 nF
WHITE_LP C6104 WHITE_AC 100 nF
WHITE_AC R6103 AGND 100 kΩ
PINK_SOURCE R6104 PINK_LP 1 kΩ
PINK_LP C6105 AGND 10 nF
PINK_LP C6106 PINK_AC 100 nF
PINK_AC R6105 AGND 100 kΩ
WHITE_SIGNAL R6106 BROWN_SUM 100 kΩ
BROWN_SHAPE R6107 BROWN_SUM 1 MΩ
BROWN_SHAPE C6107 BROWN_SUM 10 nF
PINK_SIGNAL C6108 BLUE_SERIES 100 pF
BLUE_SERIES R6108 BLUE_SUM 100 kΩ
BLUE_SHAPE R6109 BLUE_SUM 100 kΩ
BLUE_SHAPE C6109 BLUE_SUM 33 pF
WHITE_SIGNAL R6110 WHITE_LEVEL_SUM 100 kΩ
WHITE_LEVEL R6111 WHITE_TRIM 100 kΩ
WHITE_TRIM WHITE_LEVEL_SUM WHITE_LEVEL_SUM RV6101 CCW Wiper CW 10 kΩ
Cbe877b_DRIVE R6112 Cbe877b_ISO_MID 499 Ω
Cbe877b_ISO_MID R6113 WHITE_TIP 499 Ω
AGND S Tip Sleeve T WHITE_TIP TN Tip−normals
PINK_SIGNAL R6114 PINK_LEVEL_SUM 100 kΩ
PINK_LEVEL R6115 PINK_TRIM 100 kΩ
PINK_TRIM PINK_LEVEL_SUM PINK_LEVEL_SUM RV6102 CCW Wiper CW 10 kΩ
Cdbc5aa_DRIVE R6116 Cdbc5aa_ISO_MID 499 Ω
Cdbc5aa_ISO_MID R6117 PINK_TIP 499 Ω
AGND S Tip Sleeve T PINK_TIP TN Tip−normals
BLUE_SHAPE R6118 BLUE_LEVEL_SUM 10 kΩ
BLUE_LEVEL R6119 BLUE_TRIM 100 kΩ
BLUE_TRIM BLUE_LEVEL_SUM BLUE_LEVEL_SUM RV6103 CCW Wiper CW 10 kΩ
Ce44f9a_DRIVE R6120 Ce44f9a_ISO_MID 499 Ω
Ce44f9a_ISO_MID R6121 BLUE_TIP 499 Ω
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BROWN_SHAPE R6122 BROWN_LEVEL_SUM 20 kΩ
BROWN_LEVEL R6123 BROWN_TRIM 100 kΩ
BROWN_TRIM BROWN_LEVEL_SUM BROWN_LEVEL_SUM RV6104 CCW Wiper CW 10 kΩ
Cde1183_DRIVE R6124 Cde1183_ISO_MID 499 Ω
Cde1183_ISO_MID R6125 BROWN_TIP 499 Ω
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PINK_RAW PINK_SOURCE U6102B PINK_SOURCE OPA4196IDR
PINK_AC PINK_SIGNAL U6102C PINK_SIGNAL OPA4196IDR
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+12V −12V U6102E V+ V− OPA4196IDR
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AGND WHITE_LEVEL_SUM U6103B WHITE_LEVEL OPA4196IDR
AGND PINK_LEVEL_SUM U6103C PINK_LEVEL OPA4196IDR
AGND BLUE_LEVEL_SUM U6103D BLUE_LEVEL OPA4196IDR
+12V −12V U6103E V+ V− OPA4196IDR
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AGND PACK_OPA4196IDR_2_IO_UNUSED2 U6104B PACK_OPA4196IDR_2_IO_UNUSED2 OPA4196IDR
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AGND PACK_OPA4196IDR_2_IO_UNUSED4 U6104D PACK_OPA4196IDR_2_IO_UNUSED4 OPA4196IDR
+12V −12V U6104E V+ V− OPA4196IDR
WHITE_LEVEL Cbe877b_DRIVE U6105A Cbe877b_DRIVE OPA4196IDR
PINK_LEVEL Cdbc5aa_DRIVE U6105B Cdbc5aa_DRIVE OPA4196IDR
BLUE_LEVEL Ce44f9a_DRIVE U6105C Ce44f9a_DRIVE OPA4196IDR
BROWN_LEVEL Cde1183_DRIVE U6105D Cde1183_DRIVE OPA4196IDR
+12V −12V U6105E V+ V− OPA4196IDR
NOISE_VDD C6110 AGND 100 nF
+12V C6111 AGND 100 nF
−12V C6112 AGND 100 nF
+12V C6113 AGND 100 nF
−12V C6114 AGND 100 nF
+12V C6115 AGND 100 nF
−12V C6116 AGND 100 nF
+12V C6117 AGND 100 nF
−12V C6118 AGND 100 nF

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Size: A0 Date:
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Size: A0 Date: **Rev:**
KiCad E.D.A. 10.0.6 Id: 15/36

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Size: A0 Date:
Rev:
KiCad E.D.A. 10.0.6
Id: 16/36

Sheet: /E3/
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Size: A0	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 17/36

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Size: A0	Date:	**Rev:**
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Size: A0 Date: **Rev:**
KiCad E.D.A. 10.0.6 Id: 19/36

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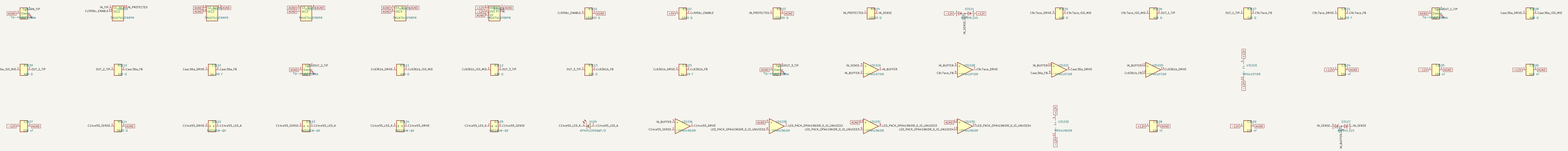
KiCad E.D.A. 10.0.6		Id: 20/36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AGND AGND S2 D2 U3101B SEL2 AGND TMUX7412FRRPR
AGND AGND S3 D3 U3101C SEL3 AGND TMUX7412FRRPR
AGND AGND S4 D4 U3101D SEL4 AGND TMUX7412FRRPR
+12V −12V AGND VDD GND VSS FF EP U3101E AGND TMUX7412FRRPR
R3101 Cc5f06c_ENABLE AGND 100000 Ω
R3102 +5V Cc5f06c_ENABLE 1000 Ω
R3103 IN_PROTECTED AGND 100000 Ω
R3104 IN_PROTECTED IN_SENSE 10000 Ω
D3101 −12V A K +12V IN_SENSE BAV199,215
R3105 C9c7ace_DRIVE C9c7ace_ISO_MID 499 Ω
R3106 C9c7ace_ISO_MID OUT_1_TIP 499 Ω
R3107 OUT_1_TIP C9c7ace_FB 100 Ω
C3101 C9c7ace_DRIVE C9c7ace_FB 1e−09 F
AGND Tip Sleeve OUT_1_TIP Tip−normal WQP518MA
R3108 Caac36a_DRIVE Caac36a_ISO_MID 499 Ω
R3109 Caac36a_ISO_MID OUT_2_TIP 499 Ω
R3110 OUT_2_TIP Caac36a_FB 100 Ω
C3102 Caac36a_DRIVE Caac36a_FB 1e−09 F
AGND Tip Sleeve OUT_2_TIP Tip−normal WQP518MA
R3111 Cc6361b_DRIVE Cc6361b_ISO_MID 499 Ω
R3112 Cc6361b_ISO_MID OUT_3_TIP 499 Ω
R3113 OUT_3_TIP Cc6361b_FB 100 Ω
C3103 Cc6361b_DRIVE Cc6361b_FB 1e−09 F
AGND Tip Sleeve OUT_3_TIP Tip−normal WQP518MA
U3102A IN_SENSE IN_BUFFER IN_BUFFER OPA4197IDR
U3102B IN_BUFFER C9c7ace_FB C9c7ace_DRIVE OPA4197IDR
U3102C IN_BUFFER Caac36a_FB Caac36a_DRIVE OPA4197IDR
U3102D IN_BUFFER Cc6361b_FB Cc6361b_DRIVE OPA4197IDR
U3102E +12V −12V V+ V− OPA4197IDR
C3104 +12V AGND 100 nF
C3105 −12V AGND 100 nF
C3106 +12V AGND 100 nF
C3107 −12V AGND 100 nF
R3114 C14ce55_SENSE AGND 5600 Ω
D3102 C14ce55_DRIVE C14ce55_LED_A BAS16GW−QX
D3103 C14ce55_SENSE C14ce55_LED_A BAS16GW−QX
D3104 C14ce55_LED_K C14ce55_DRIVE BAS16GW−QX
D3105 C14ce55_LED_K C14ce55_SENSE BAS16GW−QX
D109 C14ce55_LED_K C14ce55_LED_A APHHS1005QWF/D
U3103A IN_BUFFER C14ce55_SENSE C14ce55_DRIVE OPA4196IDR
U3103B AGND LED_PACK_OPA4196IDR_0_IO_UNUSED2 LED_PACK_OPA4196IDR_0_IO_UNUSED2 OPA4196IDR
U3103C AGND LED_PACK_OPA4196IDR_0_IO_UNUSED3 LED_PACK_OPA4196IDR_0_IO_UNUSED3 OPA4196IDR
U3103D AGND LED_PACK_OPA4196IDR_0_IO_UNUSED4 LED_PACK_OPA4196IDR_0_IO_UNUSED4 OPA4196IDR
U3103E +12V −12V V+ V− OPA4196IDR
C3108 +12V AGND 100 nF
C3109 −12V AGND 100 nF
D3107 IN_SENSE A K IN_SENSE IN_BUFFER BAT54S,215

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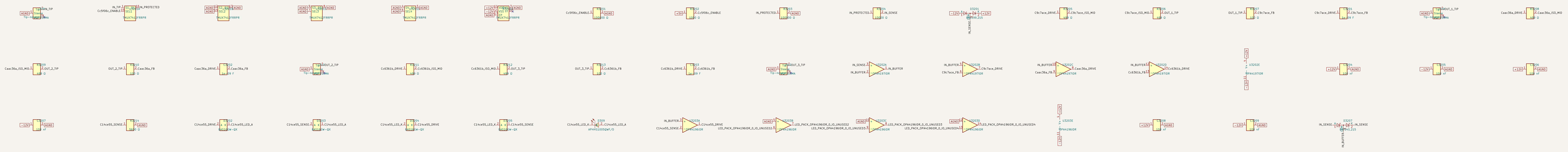
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D1
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AGND
S2
SEL2
D2
TMUX7412FRRPR
U3201C
AGND
S3
SEL3
D3
TMUX7412FRRPR
U3201D
AGND
S4
SEL4
D4
TMUX7412FRRPR
U3201E
+12V
-12V
AGND
VDD
VSS
EP
GND
FF
TMUX7412FRRPR
R3201
Cc5f06c_ENABLE
AGND
100000 Ω
R3202
+5V
Cc5f06c_ENABLE
1000 Ω
R3203
IN_PROTECTED
AGND
100000 Ω
R3204
IN_PROTECTED
IN_SENSE
10000 Ω
D3201
-12V
+12V
IN_SENSE
BAV199,215
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C9c7ace_ISO_MID
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R3206
C9c7ace_ISO_MID
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499 Ω
R3207
OUT_1_TIP
C9c7ace_FB
100 Ω
C3201
C9c7ace_DRIVE
C9c7ace_FB
1e-09 F
J409
OUT_1_TIP
AGND
R3208
Caac36a_DRIVE
Caac36a_ISO_MID
499 Ω
R3209
Caac36a_ISO_MID
OUT_2_TIP
499 Ω
R3210
OUT_2_TIP
Caac36a_FB
100 Ω
C3202
Caac36a_DRIVE
Caac36a_FB
1e-09 F
J310
OUT_2_TIP
AGND
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Cc6361b_DRIVE
Cc6361b_ISO_MID
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R3212
Cc6361b_ISO_MID
OUT_3_TIP
499 Ω
R3213
OUT_3_TIP
Cc6361b_FB
100 Ω
C3203
Cc6361b_DRIVE
Cc6361b_FB
1e-09 F
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AGND
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IN_BUFFER
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U3202B
IN_BUFFER
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C9c7ace_DRIVE
OPA4197IDR
U3202C
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Caac36a_FB
Caac36a_DRIVE
OPA4197IDR
U3202D
IN_BUFFER
Cc6361b_FB
Cc6361b_DRIVE
OPA4197IDR
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-12V
OPA4197IDR
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+12V
AGND
100 nF
C3205
-12V
AGND
100 nF
C3206
+12V
AGND
100 nF
C3207
-12V
AGND
100 nF
R3214
C14ce55_SENSE
AGND
5600 Ω
D3202
C14ce55_DRIVE
C14ce55_LED_A
BAS16GW-QX
D3203
C14ce55_SENSE
C14ce55_LED_A
BAS16GW-QX
D3204
C14ce55_LED_K
C14ce55_DRIVE
BAS16GW-QX
D3205
C14ce55_LED_K
C14ce55_SENSE
BAS16GW-QX
D309
C14ce55_LED_K
C14ce55_LED_A
APHHS1005QWF/D
U3203A
IN_BUFFER
C14ce55_SENSE
C14ce55_DRIVE
OPA4196IDR
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LED_PACK_OPA4196IDR_0_IO_UNUSED2
LED_PACK_OPA4196IDR_0_IO_UNUSED2
OPA4196IDR
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AGND
LED_PACK_OPA4196IDR_0_IO_UNUSED3
LED_PACK_OPA4196IDR_0_IO_UNUSED3
OPA4196IDR
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AGND
LED_PACK_OPA4196IDR_0_IO_UNUSED4
LED_PACK_OPA4196IDR_0_IO_UNUSED4
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C3208
+12V
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AGND
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IN_BUFFER
BAT54S,215

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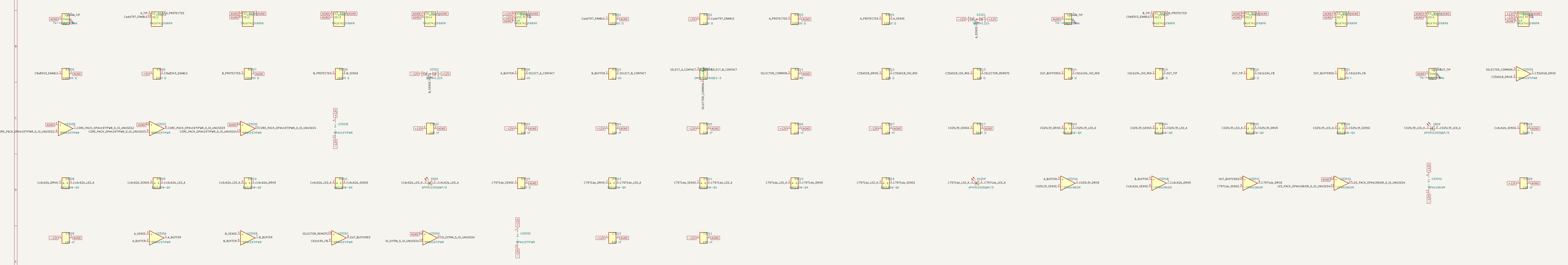
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AGND
U3301B
S2
D2
SEL2
TMUX7412FRRPR
U3301C
S3
D3
SEL3
TMUX7412FRRPR
U3301D
S4
D4
SEL4
TMUX7412FRRPR
+12V
−12V
VDD
VSS
GND
FF
EP
TMUX7412FRRPR
R3301
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100000 Ω
R3302
+5V
Cadd797_ENABLE
1000 Ω
R3303
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100000 Ω
R3304
A_PROTECTED
A_SENSE
10000 Ω
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B_TIP
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U3302C
U3302D
U3302E
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R3306
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R3308
B_PROTECTED
B_SENSE
10000 Ω
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BAT54S,215
B_SENSE
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SELECT_A_CONTACT
2.2 kΩ
R3310
B_BUFFER
SELECT_B_CONTACT
2.2 kΩ
SELECT_A_CONTACT
SELECT_B_CONTACT
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R3311
SELECTOR_COMMON
10 MΩ
R3312
C33d018_DRIVE
C33d018_ISO_MID
499 Ω
R3313
C33d018_ISO_MID
SELECTOR_REMOTE
499 Ω
R3314
OUT_BUFFERED
C61b194_ISO_MID
499 Ω
R3315
C61b194_ISO_MID
OUT_TIP
499 Ω
R3316
OUT_TIP
C61b194_FB
100 Ω
C3301
OUT_BUFFERED
C61b194_FB
1e−09 F
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C33d018_DRIVE
OPA4197IPWR
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OPA4197IPWR
U3303C
CORE_PACK_OPA4197IPWR_0_IO_UNUSED3
U3303D
CORE_PACK_OPA4197IPWR_0_IO_UNUSED4
U3303E
V+
V−
C3302
C3303
C3304
C3305
C3306
C3307
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R3317
C025c3f_SENSE
5600 Ω
D3303
C025c3f_DRIVE
C025c3f_LED_A
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C025c3f_SENSE
C025c3f_LED_A
D3305
C025c3f_LED_K
C025c3f_DRIVE
D3306
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D809
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C025c3f_LED_A
APHHS1005QWF/D
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Cc6c62b_SENSE
5600 Ω
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Cc6c62b_DRIVE
Cc6c62b_LED_A
D3309
Cc6c62b_SENSE
Cc6c62b_LED_A
D3310
Cc6c62b_LED_K
Cc6c62b_DRIVE
D3311
Cc6c62b_LED_K
Cc6c62b_SENSE
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Cc6c62b_LED_K
Cc6c62b_LED_A
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C7971de_LED_K
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D3316
C7971de_LED_K
C7971de_SENSE
D1009
C7971de_LED_K
C7971de_LED_A
APHHS1005QWF/D
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C025c3f_SENSE
C025c3f_DRIVE
OPA4196IDR
U3304B
B_BUFFER
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Cc6c62b_DRIVE
U3304C
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C7971de_DRIVE
U3304D
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U3304E
C3308
C3309
U3305A
A_SENSE
A_BUFFER
A_BUFFER
U3305B
B_SENSE
B_BUFFER
B_BUFFER
U3305C
SELECTOR_REMOTE
C61b194_FB
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U3305D
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C3311
+12V
−12V
AGND

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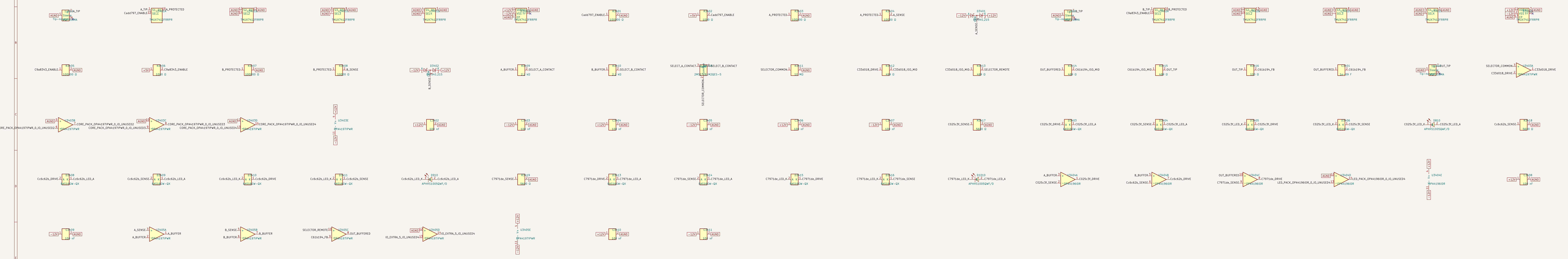
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SEL2
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AGND
U3401C
S3
D3
SEL3
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AGND
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VDD
GND
VSS
FF
EP
AGND
TMUX7412FRRPR
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R3402
Cadd797_ENABLE
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AGND
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D3401
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Sleeve
Tip-normal
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B_TIP
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D1
SEL1
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U3402B
S2
D2
SEL2
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S3
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SEL3
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D4
SEL4
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VDD
GND
VSS
FF
EP
AGND
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R3406
C9a8343_ENABLE
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R3407
AGND
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B_PROTECTED
R3408
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D3402
+12V
B_SENSE
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B_BUFFER
R3410
SELECT_B_CONTACT
2.2 kΩ
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SELECT_B_CONTACT
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SELECTOR_COMMON
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R3412
C33d018_ISO_MID
499 Ω
C33d018_ISO_MID
R3413
SELECTOR_REMOTE
499 Ω
OUT_BUFFERED
R3414
C61b194_ISO_MID
499 Ω
C61b194_ISO_MID
R3415
OUT_TIP
499 Ω
OUT_TIP
R3416
C61b194_FB
100 Ω
OUT_BUFFERED
C3401
C61b194_FB
1e-09 F
AGND
Tip
Sleeve
Tip-normal
OUT_TIP
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C33d018_DRIVE
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+12V
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AGND
100 nF
−12V
C3407
AGND
100 nF
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BAS16GW-QX
C025c3f_LED_K
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C025c3f_LED_K
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BAS16GW-QX
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C025c3f_LED_A
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Cc6c62b_LED_K
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BAS16GW-QX
Cc6c62b_LED_K
D3411
Cc6c62b_SENSE
BAS16GW-QX
Cc6c62b_LED_K
D910
Cc6c62b_LED_A
APHHS1005QWF/D
C7971de_SENSE
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5600 Ω
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C7971de_LED_K
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BAS16GW-QX
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U3404B
Cc6c62b_DRIVE
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C7971de_SENSE
U3404C
C7971de_DRIVE
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U3404D
LED_PACK_OPA4196IDR_0_IO_UNUSED4
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U3404E
OPA4196IDR
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+12V
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AGND
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−12V
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AGND
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U3405A
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B_BUFFER
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B_BUFFER
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U3405D
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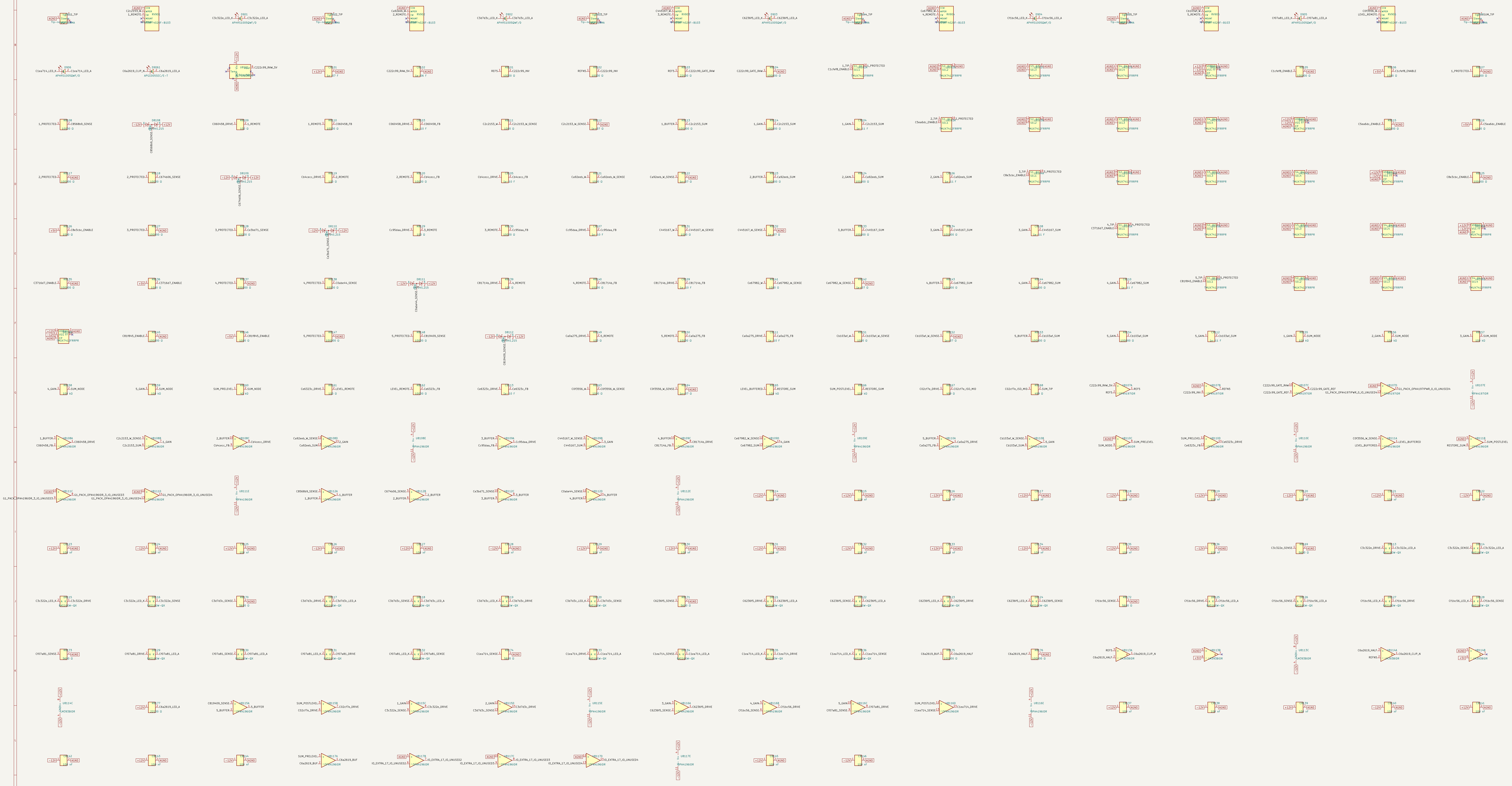

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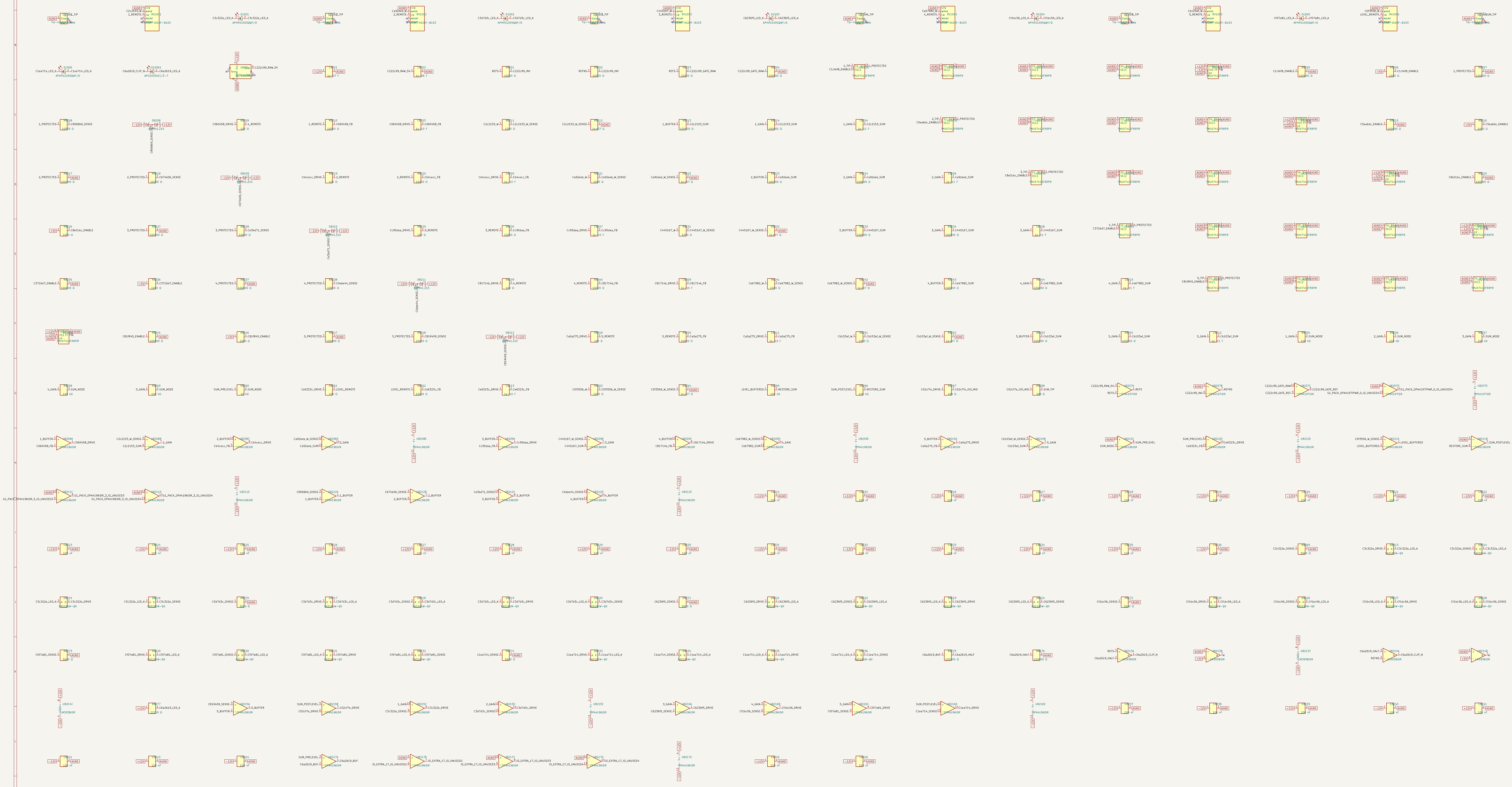

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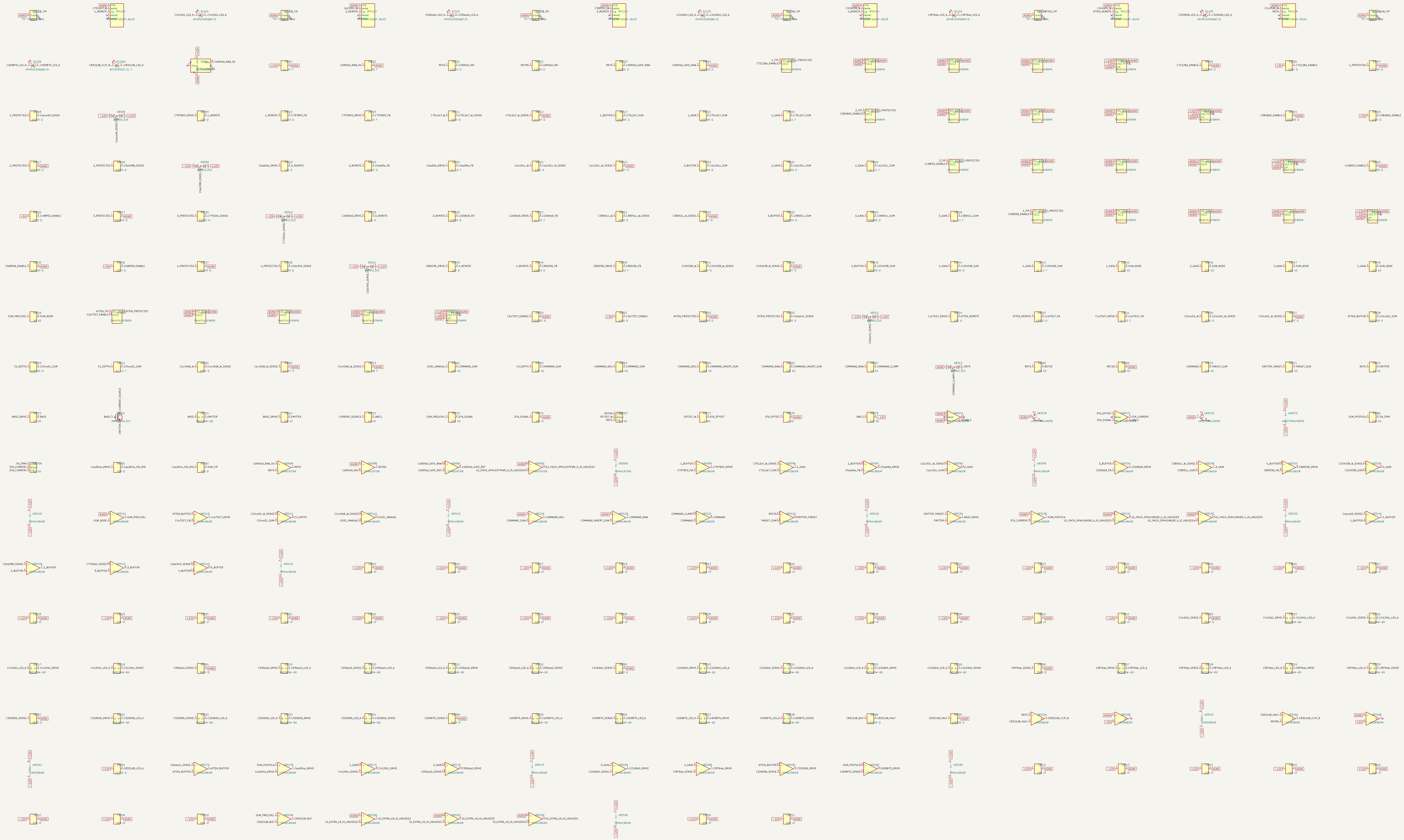

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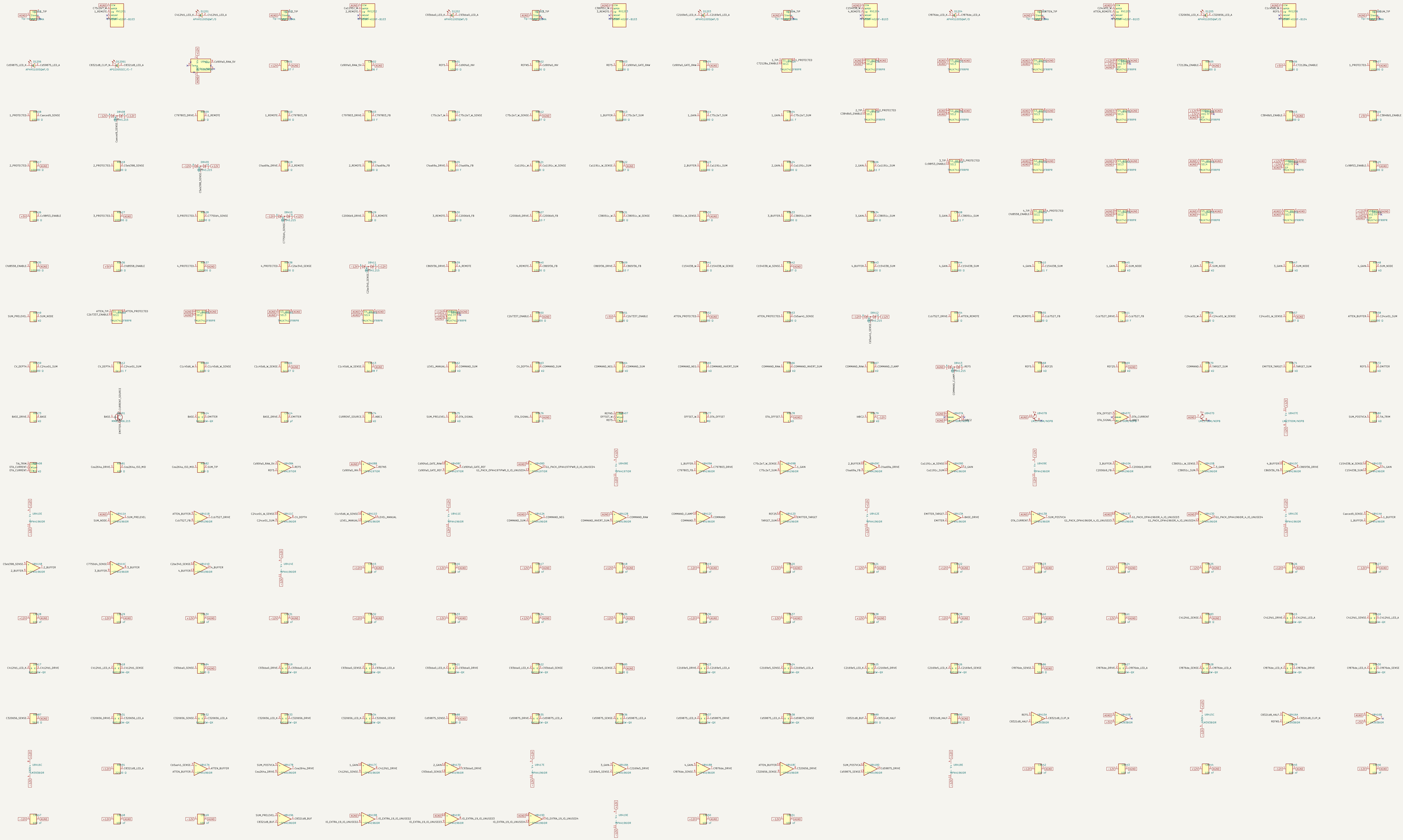

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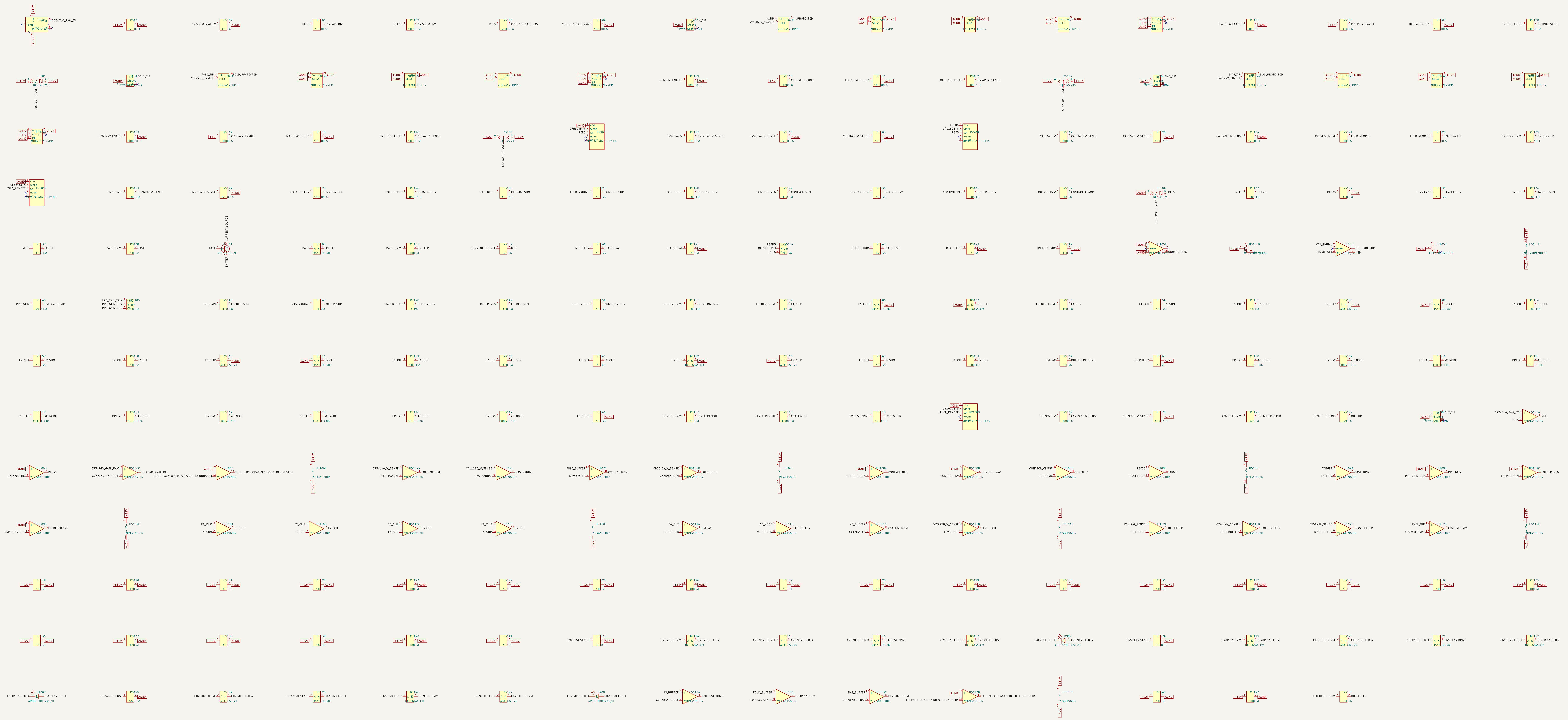

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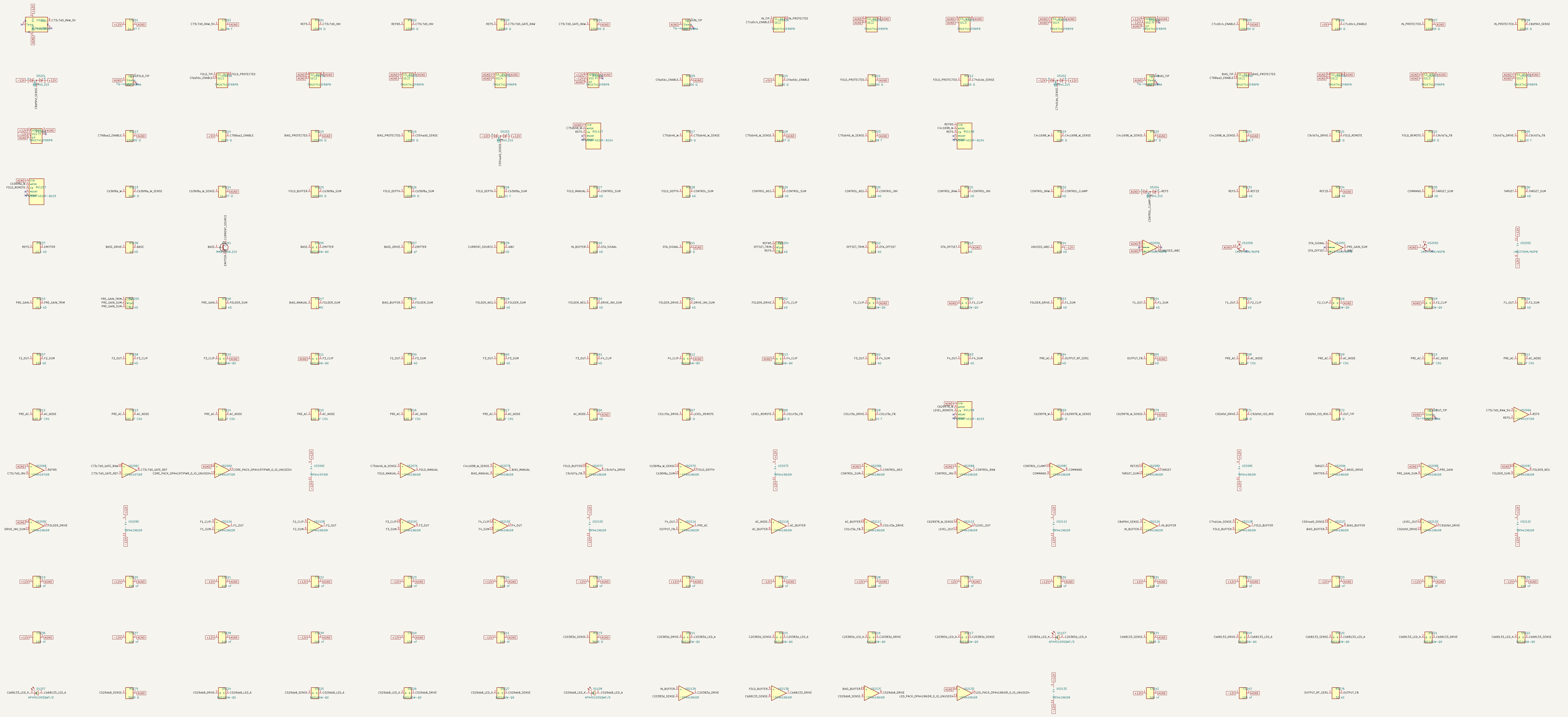

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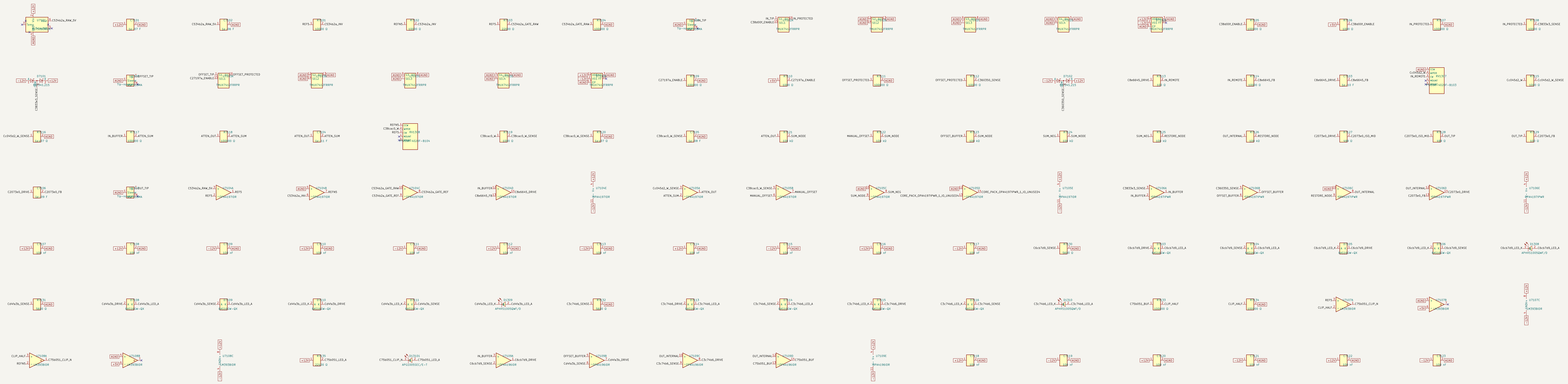

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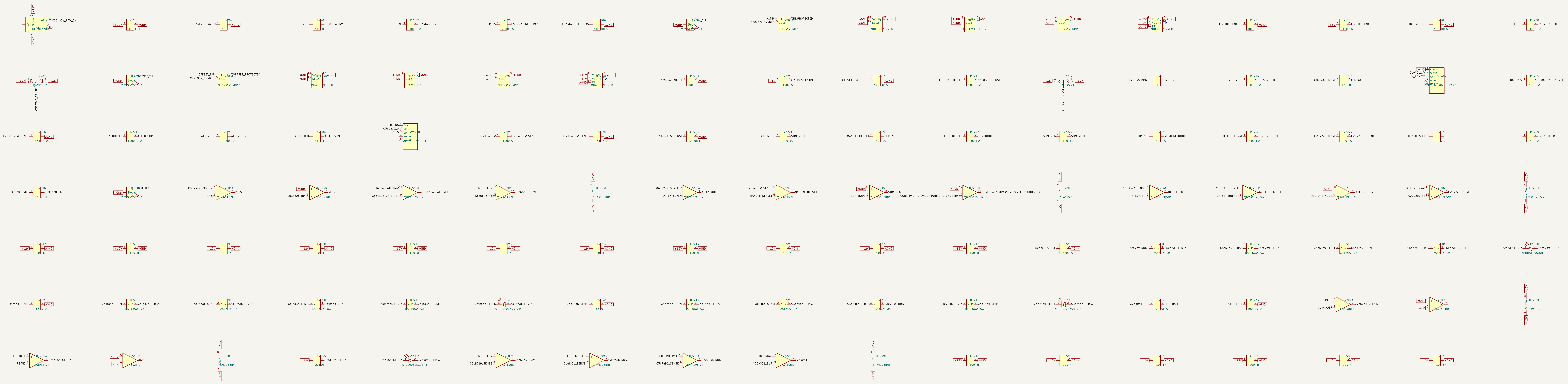

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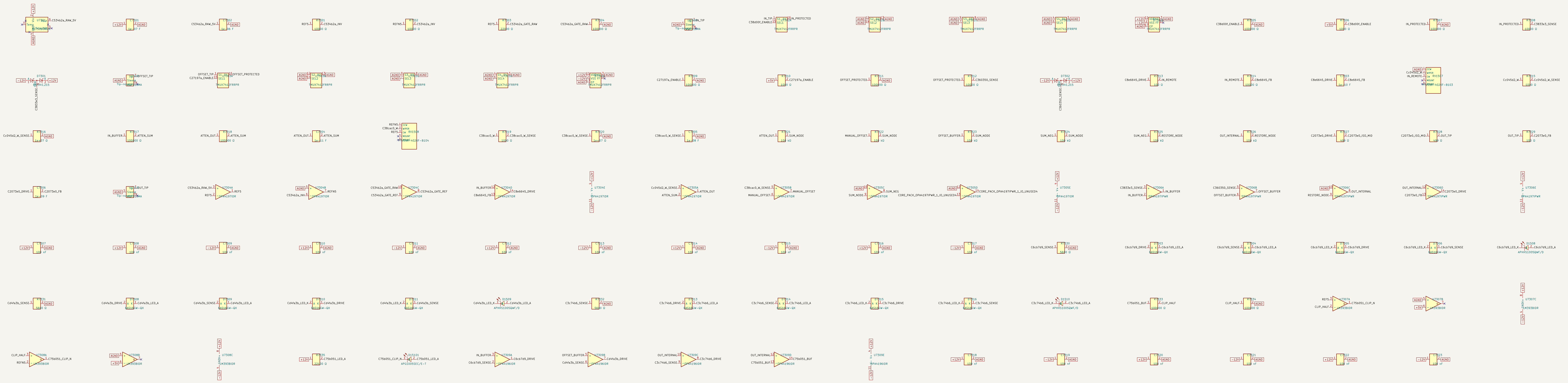
U7301 REF5050AIDR Vin Temp Vout Trim/NR GND +12V AGND C534b2a_RAW_5V
C7301 +12V AGND 1e-07 F
C7302 C534b2a_RAW_5V AGND 1e-06 F
R7301 REF5 C534b2a_INV 10000 Ω
R7302 REFN5 C534b2a_INV 10000 Ω
R7303 REF5 C534b2a_GATE_RAW 22000 Ω
R7304 C534b2a_GATE_RAW AGND 100000 Ω
AGND Tip Sleeve TN Tip-normal IN_TIP
U7302A TMUX7412FRRPR S1 D1 SEL1 IN_TIP C38d00f_ENABLE IN_PROTECTED
U7302B TMUX7412FRRPR S2 D2 SEL2 AGND
U7302C TMUX7412FRRPR S3 D3 SEL3 AGND
U7302D TMUX7412FRRPR S4 D4 SEL4 AGND
U7302E TMUX7412FRRPR VDD GND VSS FF EP +12V −12V AGND
R7305 C38d00f_ENABLE AGND 100000 Ω
R7306 +5V C38d00f_ENABLE 1000 Ω
R7307 IN_PROTECTED AGND 100000 Ω
R7308 IN_PROTECTED C3833e3_SENSE 10000 Ω
D7301 BAT54S,215 −12V +12V C3833e3_SENSE
J1509 AGND Tip Sleeve TN Tip-normal OFFSET_TIP
U7303A TMUX7412FRRPR OFFSET_TIP C27197a_ENABLE OFFSET_PROTECTED
U7303B TMUX7412FRRPR AGND
U7303C TMUX7412FRRPR AGND
U7303D TMUX7412FRRPR AGND
U7303E TMUX7412FRRPR +12V −12V AGND
R7309 C27197a_ENABLE AGND 100000 Ω
R7310 +5V C27197a_ENABLE 1000 Ω
R7311 OFFSET_PROTECTED AGND 100000 Ω
R7312 OFFSET_PROTECTED C360350_SENSE 10000 Ω
D7302 BAT54S,215 −12V +12V C360350_SENSE
R7313 C8e6645_DRIVE IN_REMOTE 100 Ω
R7314 IN_REMOTE C8e6645_FB 10000 Ω
C7303 C8e6645_DRIVE C8e6645_FB 1e-10 F
RV1507 PTV09A-4020F-B103 CCW WIPER CW MOUNT AGND Cc045d2_W IN_REMOTE
R7315 Cc045d2_W Cc045d2_W_SENSE 1000 Ω
R7316 Cc045d2_W_SENSE AGND 1e+07 Ω
R7317 IN_BUFFER ATTEN_SUM 100000 Ω
R7318 ATTEN_OUT ATTEN_SUM 100000 Ω
C7304 ATTEN_OUT ATTEN_SUM 1e-11 F
RV1508 PTV09A-4020F-B104 CCW WIPER CW MOUNT REFN5 C38cac0_W REF5
R7319 C38cac0_W C38cac0_W_SENSE 1000 Ω
R7320 C38cac0_W_SENSE AGND 1e+07 Ω
C7305 C38cac0_W_SENSE AGND 1e-08 F
R7321 ATTEN_OUT SUM_NODE 100 kΩ
R7322 MANUAL_OFFSET SUM_NODE 100 kΩ
R7323 OFFSET_BUFFER SUM_NODE 100 kΩ
R7324 SUM_NEG SUM_NODE 100 kΩ
R7325 SUM_NEG RESTORE_NODE 100 kΩ
R7326 OUT_INTERNAL RESTORE_NODE 100 kΩ
R7327 C2073e0_DRIVE C2073e0_ISO_MID 499 Ω
R7328 C2073e0_ISO_MID OUT_TIP 499 Ω
R7329 OUT_TIP C2073e0_FB 100 Ω
C7306 C2073e0_DRIVE C2073e0_FB 1e-09 F
J1510 AGND Tip Sleeve TN Tip-normal OUT_TIP
U7304A OPA4197IDR C534b2a_RAW_5V REF5 REF5
U7304B OPA4197IDR AGND C534b2a_INV REFN5
U7304C OPA4197IDR C534b2a_GATE_RAW C534b2a_GATE_REF C534b2a_GATE_REF
U7304D OPA4197IDR IN_BUFFER C8e6645_FB C8e6645_DRIVE
U7304E OPA4197IDR +12V −12V
U7305A OPA4197IDR Cc045d2_W_SENSE ATTEN_SUM ATTEN_OUT
U7305B OPA4197IDR C38cac0_W_SENSE MANUAL_OFFSET MANUAL_OFFSET
U7305C OPA4197IDR AGND SUM_NODE SUM_NEG
U7305D OPA4197IDR AGND CORE_PACK_OPA4197IPWR_1_IO_UNUSED4 CORE_PACK_OPA4197IPWR_1_IO_UNUSED4
U7305E OPA4197IDR +12V −12V
U7306A OPA4197IPWR C3833e3_SENSE IN_BUFFER IN_BUFFER
U7306B OPA4197IPWR C360350_SENSE OFFSET_BUFFER OFFSET_BUFFER
U7306C OPA4197IPWR AGND RESTORE_NODE OUT_INTERNAL
U7306D OPA4197IPWR OUT_INTERNAL C2073e0_FB C2073e0_DRIVE
U7306E OPA4197IPWR +12V −12V
C7307 +12V AGND 100 nF
C7308 +12V AGND 100 nF
C7309 −12V AGND 100 nF
C7310 +12V AGND 100 nF
C7311 −12V AGND 100 nF
C7312 +12V AGND 100 nF
C7313 −12V AGND 100 nF
C7314 +12V AGND 100 nF
C7315 −12V AGND 100 nF
C7316 +12V AGND 100 nF
C7317 −12V AGND 100 nF
R7330 C6cb7d9_SENSE AGND 5600 Ω
D7303 BAS16GW-QX C6cb7d9_DRIVE C6cb7d9_LED_A
D7304 BAS16GW-QX C6cb7d9_SENSE C6cb7d9_LED_A
D7305 BAS16GW-QX C6cb7d9_LED_K C6cb7d9_DRIVE
D7306 BAS16GW-QX C6cb7d9_LED_K C6cb7d9_SENSE
D1508 APHHS1005QWF/D C6cb7d9_LED_K C6cb7d9_LED_A
R7331 Cd4fa3b_SENSE AGND 5600 Ω
D7308 BAS16GW-QX Cd4fa3b_DRIVE Cd4fa3b_LED_A
D7309 BAS16GW-QX Cd4fa3b_SENSE Cd4fa3b_LED_A
D7310 BAS16GW-QX Cd4fa3b_LED_K Cd4fa3b_DRIVE
D7311 BAS16GW-QX Cd4fa3b_LED_K Cd4fa3b_SENSE
D1509 APHHS1005QWF/D Cd4fa3b_LED_K Cd4fa3b_LED_A
R7332 C3c74b6_SENSE AGND 5600 Ω
D7313 BAS16GW-QX C3c74b6_DRIVE C3c74b6_LED_A
D7314 BAS16GW-QX C3c74b6_SENSE C3c74b6_LED_A
D7315 BAS16GW-QX C3c74b6_LED_K C3c74b6_DRIVE
D7316 BAS16GW-QX C3c74b6_LED_K C3c74b6_SENSE
D1510 APHHS1005QWF/D C3c74b6_LED_K C3c74b6_LED_A
R7333 C75b051_BUF CLIP_HALF 100000 Ω
R7334 CLIP_HALF AGND 100000 Ω
U7307A LM393BIDR REF5 CLIP_HALF C75b051_CLIP_N
U7307B LM393BIDR AGND +5V
U7307C LM393BIDR GND V+ +12V −12V
U7308A LM393BIDR CLIP_HALF REFN5 C75b051_CLIP_N
U7308B LM393BIDR AGND +5V
U7308C LM393BIDR GND V+ +12V −12V
R7335 +12V C75b051_LED_A 22100 Ω
D15101 APG1005SEC/E-T C75b051_CLIP_N C75b051_LED_A
U7309A OPA4196IDR IN_BUFFER C6cb7d9_SENSE C6cb7d9_DRIVE
U7309B OPA4196IDR OFFSET_BUFFER Cd4fa3b_SENSE Cd4fa3b_DRIVE
U7309C OPA4196IDR OUT_INTERNAL C3c74b6_SENSE C3c74b6_DRIVE
U7309D OPA4196IDR OUT_INTERNAL C75b051_BUF C75b051_BUF
U7309E OPA4196IDR +12V −12V
C7318 +12V AGND 100 nF
C7319 −12V AGND 100 nF
C7320 +12V AGND 100 nF
C7321 −12V AGND 100 nF
C7322 +12V AGND 100 nF
C7323 −12V AGND 100 nF

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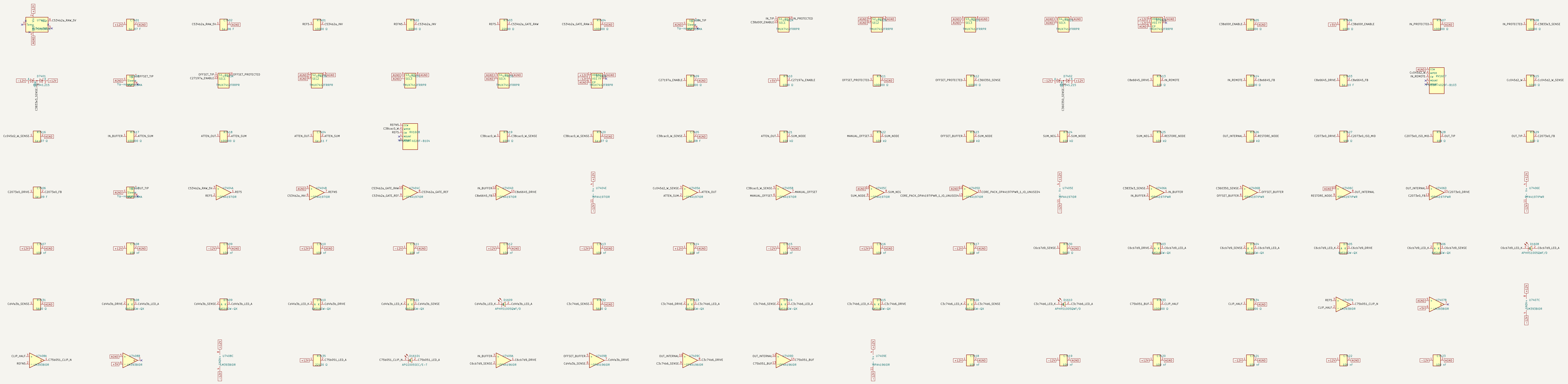
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Vout
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AGND
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C7401
1e-07 F
C7402
1e-06 F
R7401
REF5
C534b2a_INV
10000 Ω
R7402
REFN5
10000 Ω
R7403
C534b2a_GATE_RAW
22000 Ω
R7404
100000 Ω
J1609
Tip
Sleeve
Tip-normal
IN_TIP
U7402A
S1
D1
SEL1
IN_PROTECTED
C38d00f_ENABLE
TMUX7412FRRPR
U7402B
S2
D2
SEL2
U7402C
S3
D3
SEL3
U7402D
S4
D4
SEL4
U7402E
VDD
VSS
GND
EP
FF
-12V
R7405
R7406
+5V
C38d00f_ENABLE
1000 Ω
R7407
R7408
C3833e3_SENSE
D7401
BAT54S,215
C3833e3_SENSE
J1610
OFFSET_TIP
U7403A
OFFSET_PROTECTED
C27197a_ENABLE
U7403B
U7403C
U7403D
U7403E
R7409
R7410
R7411
R7412
C360350_SENSE
D7402
R7413
C8e6645_DRIVE
IN_REMOTE
100 Ω
R7414
C8e6645_FB
C7403
1e-10 F
RV1607
CCW
WIPER
CW
MOUNT
Cc045d2_W
PTV09A-4020F-B103
R7415
Cc045d2_W_SENSE
R7416
1e+07 Ω
R7417
IN_BUFFER
ATTEN_SUM
R7418
ATTEN_OUT
C7404
1e-11 F
RV1608
C38cac0_W
PTV09A-4020F-B104
R7419
C38cac0_W_SENSE
R7420
C7405
1e-08 F
R7421
SUM_NODE
100 kΩ
R7422
MANUAL_OFFSET
R7423
OFFSET_BUFFER
R7424
SUM_NEG
R7425
RESTORE_NODE
R7426
OUT_INTERNAL
R7427
C2073e0_DRIVE
C2073e0_ISO_MID
499 Ω
R7428
OUT_TIP
R7429
C2073e0_FB
C7406
1e-09 F
J1611
U7404A
OPA4197IDR
U7404B
U7404C
C534b2a_GATE_REF
U7404D
U7404E
V+
V-
U7405A
U7405B
U7405C
U7405D
CORE_PACK_OPA4197IPWR_1_IO_UNUSED4
U7405E
U7406A
OPA4197IPWR
U7406B
U7406C
U7406D
U7406E
C7407
100 nF
C7408
C7409
C7410
C7411
C7412
C7413
C7414
C7415
C7416
C7417
R7430
C6cb7d9_SENSE
5600 Ω
D7403
A
K
BAS16GW-QX
C6cb7d9_DRIVE
C6cb7d9_LED_A
D7404
D7405
C6cb7d9_LED_K
D7406
D1608
APHHS1005QWF/D
R7431
Cd4fa3b_SENSE
D7408
Cd4fa3b_DRIVE
Cd4fa3b_LED_A
D7409
D7410
Cd4fa3b_LED_K
D7411
D1609
R7432
C3c74b6_SENSE
D7413
C3c74b6_DRIVE
C3c74b6_LED_A
D7414
D7415
C3c74b6_LED_K
D7416
D1610
R7433
C75b051_BUF
CLIP_HALF
R7434
U7407A
C75b051_CLIP_N
LM393BIDR
U7407B
U7407C
GND
U7408A
U7408B
U7408C
R7435
C75b051_LED_A
22100 Ω
D16101
APG1005SEC/E-T
U7409A
OPA4196IDR
U7409B
U7409C
U7409D
U7409E
C7418
C7419
C7420
C7421
C7422
C7423

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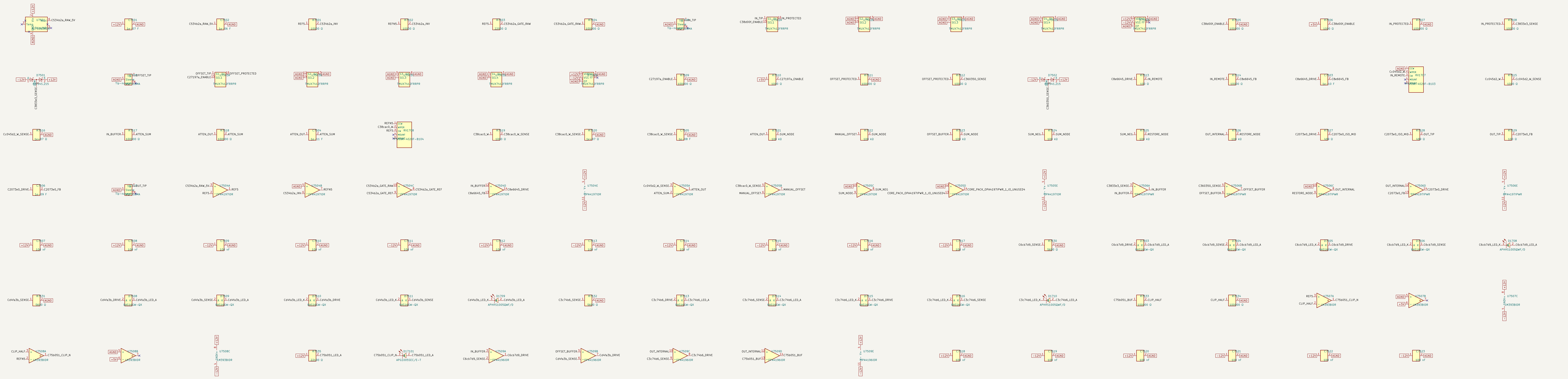
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Vout
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GND
Trim/NR
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+12V
AGND
C7501
1e-07 F
C7502
1e-06 F
R7501
REF5
C534b2a_INV
10000 Ω
R7502
REFN5
D7501
BAT54S,215
C3833e3_SENSE
-12V
OFFSET_TIP
OFFSET_PROTECTED
C27197a_ENABLE
U7503A
SEL1
TMUX7412FRRPR
U7503B
SEL2
U7503C
SEL3
R7516
Cc045d2_W_SENSE
1e+07 Ω
R7517
IN_BUFFER
ATTEN_SUM
100000 Ω
R7518
ATTEN_OUT
C7504
1e-11 F
RV1708
CCW
WIPER
CW
MOUNT
C38cac0_W
C7506
C2073e0_DRIVE
C2073e0_FB
1e-09 F
OUT_TIP
Tip
Sleeve
U7504A
U7504B
U7504C
OPA4197IDR
C534b2a_GATE_RAW
C534b2a_GATE_REF

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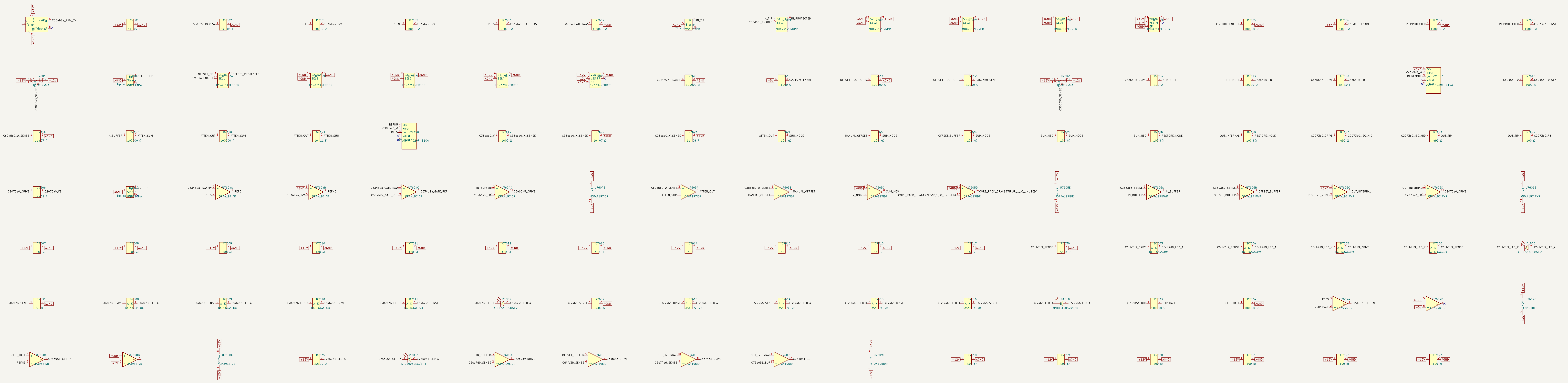

U7601 REF5050IDR Vin Vout Temp GND +12V AGND C534b2a_RAW_5V
C7601 1e-07 F +12V AGND
C7602 1e-06 F C534b2a_RAW_5V AGND
R7601 10000 Ω REF5 C534b2a_INV
R7602 10000 Ω REFN5 C534b2a_INV
R7603 22000 Ω REF5 C534b2a_GATE_RAW
R7604 100000 Ω C534b2a_GATE_RAW AGND
Tip Sleeve Tip-normal AGND IN_TIP
U7602A TMUX7412FRRPR S1 D1 SEL1 IN_TIP C38d00f_ENABLE IN_PROTECTED
U7602B TMUX7412FRRPR S2 D2 SEL2 AGND
U7602C TMUX7412FRRPR S3 D3 SEL3 AGND
U7602D TMUX7412FRRPR S4 D4 SEL4 AGND
U7602E TMUX7412FRRPR VDD VSS EP +12V −12V AGND
R7605 100000 Ω C38d00f_ENABLE AGND
R7606 1000 Ω +5V C38d00f_ENABLE
R7607 100000 Ω IN_PROTECTED AGND
R7608 10000 Ω IN_PROTECTED C3833e3_SENSE
D7601 BAT54S,215 −12V +12V C3833e3_SENSE
Tip Sleeve Tip-normal AGND OFFSET_TIP
U7603A TMUX7412FRRPR S1 D1 SEL1 OFFSET_TIP C27197a_ENABLE OFFSET_PROTECTED
U7603B TMUX7412FRRPR SEL2 AGND
U7603C TMUX7412FRRPR SEL3 AGND
U7603D TMUX7412FRRPR SEL4 AGND
U7603E TMUX7412FRRPR VDD VSS EP +12V −12V AGND
R7609 100000 Ω C27197a_ENABLE AGND
R7610 1000 Ω +5V C27197a_ENABLE
R7611 100000 Ω OFFSET_PROTECTED AGND
R7612 10000 Ω OFFSET_PROTECTED C360350_SENSE
D7602 BAT54S,215 −12V +12V C360350_SENSE
R7613 100 Ω C8e6645_DRIVE IN_REMOTE
R7614 10000 Ω IN_REMOTE C8e6645_FB
C7603 1e-10 F C8e6645_DRIVE C8e6645_FB
RV1807 PTV09A-4020F-B103 CCW WIPER CW MOUNT AGND Cc045d2_W IN_REMOTE
R7615 1000 Ω Cc045d2_W Cc045d2_W_SENSE
R7616 1e+07 Ω Cc045d2_W_SENSE AGND
R7617 100000 Ω IN_BUFFER ATTEN_SUM
R7618 100000 Ω ATTEN_OUT ATTEN_SUM
C7604 1e-11 F ATTEN_OUT ATTEN_SUM
RV1808 PTV09A-4020F-B104 CCW WIPER CW MOUNT REFN5 C38cac0_W REF5
R7619 1000 Ω C38cac0_W C38cac0_W_SENSE
R7620 1e+07 Ω C38cac0_W_SENSE AGND
C7605 1e-08 F C38cac0_W_SENSE AGND
R7621 100 kΩ ATTEN_OUT SUM_NODE
R7622 100 kΩ MANUAL_OFFSET SUM_NODE
R7623 100 kΩ OFFSET_BUFFER SUM_NODE
R7624 100 kΩ SUM_NEG SUM_NODE
R7625 100 kΩ SUM_NEG RESTORE_NODE
R7626 100 kΩ OUT_INTERNAL RESTORE_NODE
R7627 499 Ω C2073e0_DRIVE C2073e0_ISO_MID
R7628 499 Ω C2073e0_ISO_MID OUT_TIP
R7629 100 Ω OUT_TIP C2073e0_FB
C7606 1e-09 F C2073e0_DRIVE C2073e0_FB
Tip Sleeve Tip-normal AGND OUT_TIP
U7604A OPA4197IDR C534b2a_RAW_5V REF5
U7604B OPA4197IDR AGND C534b2a_INV REFN5
U7604C OPA4197IDR C534b2a_GATE_RAW C534b2a_GATE_REF
U7604D OPA4197IDR IN_BUFFER C8e6645_FB C8e6645_DRIVE
U7604E OPA4197IDR V+ V− +12V −12V
U7605A OPA4197IDR Cc045d2_W_SENSE ATTEN_SUM ATTEN_OUT
U7605B OPA4197IDR C38cac0_W_SENSE MANUAL_OFFSET
U7605C OPA4197IDR AGND SUM_NODE SUM_NEG
U7605D OPA4197IDR AGND CORE_PACK_OPA4197IPWR_1_IO_UNUSED4
U7605E OPA4197IDR V+ V− +12V −12V
U7606A OPA4197IPWR C3833e3_SENSE IN_BUFFER
U7606B OPA4197IPWR C360350_SENSE OFFSET_BUFFER
U7606C OPA4197IPWR AGND RESTORE_NODE OUT_INTERNAL
U7606D OPA4197IPWR OUT_INTERNAL C2073e0_FB C2073e0_DRIVE
U7606E OPA4197IPWR V+ V− +12V −12V
C7607 100 nF +12V AGND
C7608 100 nF +12V AGND
C7609 100 nF −12V AGND
C7610 100 nF +12V AGND
C7611 100 nF −12V AGND
C7612 100 nF +12V AGND
C7613 100 nF −12V AGND
C7614 100 nF +12V AGND
C7615 100 nF −12V AGND
C7616 100 nF +12V AGND
C7617 100 nF −12V AGND
R7630 5600 Ω C6cb7d9_SENSE AGND
D7603 BAS16GW-QX C6cb7d9_DRIVE C6cb7d9_LED_A
D7604 BAS16GW-QX C6cb7d9_SENSE C6cb7d9_LED_A
D7605 BAS16GW-QX C6cb7d9_LED_K C6cb7d9_DRIVE
D7606 BAS16GW-QX C6cb7d9_LED_K C6cb7d9_SENSE
D1808 APHHS1005QWF/D C6cb7d9_LED_K C6cb7d9_LED_A
R7631 5600 Ω Cd4fa3b_SENSE AGND
D7608 BAS16GW-QX Cd4fa3b_DRIVE Cd4fa3b_LED_A
D7609 BAS16GW-QX Cd4fa3b_SENSE Cd4fa3b_LED_A
D7610 BAS16GW-QX Cd4fa3b_LED_K Cd4fa3b_DRIVE
D7611 BAS16GW-QX Cd4fa3b_LED_K Cd4fa3b_SENSE
D1809 APHHS1005QWF/D Cd4fa3b_LED_K Cd4fa3b_LED_A
R7632 5600 Ω C3c74b6_SENSE AGND
D7613 BAS16GW-QX C3c74b6_DRIVE C3c74b6_LED_A
D7614 BAS16GW-QX C3c74b6_SENSE C3c74b6_LED_A
D7615 BAS16GW-QX C3c74b6_LED_K C3c74b6_DRIVE
D7616 BAS16GW-QX C3c74b6_LED_K C3c74b6_SENSE
D1810 APHHS1005QWF/D C3c74b6_LED_K C3c74b6_LED_A
R7633 100000 Ω C75b051_BUF CLIP_HALF
R7634 100000 Ω CLIP_HALF AGND
U7607A LM393BIDR REF5 CLIP_HALF C75b051_CLIP_N
U7607B LM393BIDR AGND +5V
U7607C LM393BIDR V+ GND +12V −12V
U7608A LM393BIDR CLIP_HALF REFN5 C75b051_CLIP_N
U7608B LM393BIDR AGND +5V
U7608C LM393BIDR V+ GND +12V −12V
R7635 22100 Ω +12V C75b051_LED_A
D18101 APG1005SEC/E-T C75b051_CLIP_N C75b051_LED_A
U7609A OPA4196IDR IN_BUFFER C6cb7d9_SENSE C6cb7d9_DRIVE
U7609B OPA4196IDR OFFSET_BUFFER Cd4fa3b_SENSE Cd4fa3b_DRIVE
U7609C OPA4196IDR OUT_INTERNAL C3c74b6_SENSE C3c74b6_DRIVE
U7609D OPA4196IDR OUT_INTERNAL C75b051_BUF
U7609E OPA4196IDR V+ V− +12V −12V
C7618 100 nF +12V AGND
C7619 100 nF −12V AGND
C7620 100 nF +12V AGND
C7621 100 nF −12V AGND
C7622 100 nF +12V AGND
C7623 100 nF −12V AGND

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